does not have any meaning even though the two statements about

$$\lim_{x\to x_0} f(x) \text{ and } \lim_{x\to x_0} g(x)$$

may have.

[105]Exercise 5.2.11. What exactly is the domain of the function $f(x)/g(x)$? Show that x_0 would be a point of accumulation of that domain provided that $g(x) \to C$ as $x \to x_0$ and $C \neq 0$.

[106]Exercise 5.2.28. It is enough to assume that $\lim_{x\to x_0} f(x)$ exists and to apply Theorem 5.25 with $F(x) = |x|$. Be sure to explain why this function F has the properties expressed in that theorem.

[107]Exercise 5.2.29. It is enough to assume that $\lim_{x\to x_0} f(x)$ exists and is positive and then apply Theorem 5.25 with $F(x) = \sqrt{x}$. Alternatively, assume that $f(x) \geq 0$ for all x in a neighborhood of x_0. Again be sure to explain why this function F has the properties expressed in that theorem.

[108]Exercise 5.2.32. Use the property of exponentials that $e^{a+b} = e^a e^b$ and the product rule for limits.

[109]Exercise 5.2.33. Use a trigonometric identity for $\sin(x - x_0 + x_0)$ and the sum and products rule for limits.

[110]Exercise 5.2.34. Take the function $H(x)$ of the text and consider instead $H(x) + x$.

[111]Exercise 5.2.36. This would be trivial if the sets A_i were disjoint. So it is the case where these are not disjoint that you need to address.

[112]Exercise 5.2.44. If x_0 is not in the Cantor set K, then it is in some open interval complementary to that set. Use that to prove the existence of the limit. If x_0 is in the Cantor set, then there must be sequences $x_n \to x_0$ and $y_n \to x_0$ with $x_n \in K$ and $y_n \notin K$. Use that to prove the nonexistence of the limit.

[113]Exercise 5.3.5. Consider separately the cases $x_0 \in E$ and $x_0 \notin E$. Under what circumstances in the latter case would the lim sup be larger according to this revised definition?

[114]Exercise 5.4.15. One of the definitions treats isolated points in a special way. Note that each point in the domain of f is isolated.

[115]Exercise 5.4.17. You must arrange for $f(0)$ to be the limit of the sequence of values $f(2^{-n})$. No other condition is necessary.

[116]Exercise 5.4.19. At an isolated point x_0 of the domain the limit $\lim_{x\to x_0} f(x)$ has no meaning. But if x_0 is not an isolated point in the domain of f it must be a point of accumulation and then $\lim_{x\to x_0} f(x)$ is defined and it must be equal to $f(x_0)$.

[117]Exercise 5.4.20. For the converse consider the function $f(x) = \sqrt{x}$ on $[0,1]$.

[118]Exercise 5.6.1. Let $a = \inf K$ and $b = \sup K$ and apply Cousin's lemma to the interval $[a,b]$ by taking the same collection nearly, namely $\mathcal{C}$ consist of all closed subintervals $[t,s]$ such that

$$|f(t') - f(s')| < \varepsilon/2$$

for all $t', s' \in K \cap [t,s]$. You will have to find a different choice of δ to make your argument work.

[119]Exercise 5.6.2. As usual in applications of Cousin's lemma, we should define first our collection of closed subintervals so as to have a desired property that can be extended to the whole interval $[a,b]$. Let $\varepsilon > 0$. Let $\mathcal{C}$ consist of all closed subintervals $[t,s]$ such that

$$|f(t') - f(s')| < \varepsilon/2$$

for all $t', s' \in [t,s]$. We check that $\mathcal{C}$ satisfies the hypotheses of Lemma 4.26.

For each $x \in [a,b]$ there exists $\delta(x) > 0$ such that if

$$t \in [a,b] \cap (x-\delta(x), x+\delta(x)),$$

then

$$|f(t) - f(x)| < \varepsilon/4.$$

It follows that if t' and s' are in the set

$$[a,b] \cap (x-\delta(x), x+\delta(x)),$$

then

$$\begin{aligned} |f(t') - f(s')| &\leq |f(t') - f(x)| + |f(x) - f(s')| \\ &< \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}. \end{aligned}$$

Consequently, every interval $[t,s]$ inside

$$[a,b] \cap (x-\delta(x), x+\delta(x))$$

belongs to $\mathcal{C}$.

Thus Lemma 4.26 may be applied and there exists a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

such that if, for some $i = 1, \ldots, n$,

$$x_{i-1} \leq x, y \leq x_i,$$

then

$$|f(x) - f(y)| < \varepsilon/2.$$

Let

$$\delta = \min_{i=1,\ldots,n} |x_i - x_{i-1}|.$$

If $x < y$ and $|x-y| < \delta$, then either there exists i for which

$$x_{i-1} \leq x, y \leq x_i,$$

in which case

$$|f(x) - f(y) < \varepsilon/2,$$

or there exists i such that

$$x_{i-1} \leq x \leq x_i \leq y \leq x_{i+1},$$

in which case

$$\begin{aligned} |f(y) - f(x)| &\leq |f(y) - f(x_i)| \\ &+ |f(x_i) - f(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Since this argument applies to any positive ε, we have proved that f is uniformly continuous on $[a,b]$.

[120]Exercise 5.6.5. If X is compact (closed and bounded) then this property should hold. If the set X has a point of accumulation that does not belong to X, then it is possible to give an example of a continuous function defined on X that is not uniformly continuous on X. Finally consider the situation in which the set is closed but not bounded: are there points x_i and x_j arbitrarily close together?

[121]Exercise 5.6.7. You need consider only two compact sets X_1, X_2. Since they are compact, there is a positive distance between them that you can use to help define your δ. For not closed consider $X_1 = (0,1)$ and $X_2 = (1,2)$ and define f appropriately. For not bounded use

$$X_1 = \{1, 2, 3, \ldots\}$$

and

$$X_2 = \{1, 2+1/2, 3+1/3, 4+1/4, \ldots$$

and define f appropriately.

[122]Exercise 5.6.9. For the converse consider the function $f(x) = \sqrt{x}$ on $[0,1]$. By Theorem 5.48 we know that this function is uniformly continuous on $[0,1]$.

[123]Exercise 5.6.10. Show that any function defined on a set X containing just one element is uniformly continuous. Then consider the sequence $X_i = \{x_i\}$, $i = 1, 2, \ldots, n$.

[124]Exercise 5.6.11. For the sequence of intervals you might choose $[1,2]$, $[2,3]$, $[3,4]$, ... (Why would you not be able to choose $[1/2,1]$, $[1/4,1/2]$, $[1/8,1/4]$, ...?)

[125]Exercise 5.6.12. This can be obtained merely by negating the formal statement that f is uniformly continuous on $[a,b]$.

[126]Exercise 5.6.13. Using the local continuity property, claim that there are open intervals I_x containing any point x so that

$$|f(y) - f(x)| < \varepsilon$$

for any $y \in I_x$. Now apply the Heine-Borel property to this open cover. Obtain uniform continuity from the finite subcover.

[127]Exercise 5.6.15. Let $\mathcal{C}$ be the collection of all closed intervals $I \subset [a,b]$ so that f is bounded on I. Use Cousin's lemma to find a partition of $[a,b]$ using intervals in $\mathcal{C}$.

[128]Exercise 5.6.16. Use an indirect proof. Show that if f is not bounded then there is a sequence $\{x_n\}$ of points in $[a,b]$ so that

$$|f(x_n)| > n$$

for all n. Now apply the Bolzano-Weierstrass property to obtain subsequences and get a contradiction.

[129]Exercise 5.6.17. Using the local continuity property, claim that there are open intervals I_x containing any point x so that

$$|f(y) - f(x)| < 1$$

for any $y \in I_x$. Now apply the Heine-Borel property to this open cover. Obtain boundedness of f from the finite subcover.

[130]Exercise 5.7.2. That is, prove that the image set

$$f(K) = \{f(x) : x \in K\}$$

is compact if K is compact and f is a continuous function defined at every point of K. Give a direct proof that uses the fact that a set is compact if and only if every sequence in the set has a subsequence convergent to a point in the set. Start with a sequence of points $\{y_n\}$ in $f(K)$, explain why there must be a sequence $\{x_n\}$ in K with $f(x_n) = y_n$ etc.

[131]Exercise 5.7.3. Let

$$M = \sup\{f(x) : a \le x \le b\}.$$

Explain why you can choose a sequence of points $\{x_n\}$ from $[a,b]$ so that

$$f(x_n) > M - 1/n.$$

Now apply the Bolzano-Weierstrass theorem and use the continuity of f.

[132]Exercise 5.7.5. If $f(x_0) = c > 0$, then there is an interval $[-N,N]$ so that $x_0 \in [-N,N]$ and $|f(x)| < c/2$ for all $x > N$ and $x < -N$.

[133]Exercise 5.8.2. That is, prove that the image set $f([c,d])$ is a compact interval for any interval $[c,d]$ if f is a continuous function defined at every point of $[c,d]$. Apply Theorem 5.52 and Theorem 5.53.

[134]Exercise 5.8.3. Suppose that the theorem is false and explain, then, why there should exist sequences $\{x_n\}$ and $\{y_n\}$ from $[a,b]$ so that $f(x_n) > c$, $f(y_n) < c$ and $|x_n - y_n| < 1/n$.

[135]Exercise 5.8.4. Suppose that the theorem is false and explain, then, why there should exist at each point $x \in [a,b]$ an open interval I_x centered at x so that either $f(t) > c$ for all $t \in I_x \cap [a,b]$ or else $f(t) < c$ for all $t \in I_x \cap [a,b]$.

[136]Exercise 5.8.5. You may take $c = 0$. Show that if $f(z) > 0$, then there is an interval $[z-\delta, z]$ on which f is positive. Show that if $f(z) < 0$, then there is an interval $[z, z+\delta]$ on which f is negative. Explain why each of these two cases is impossible.

[137]Exercise 5.8.6. The function must be onto. Hence there is a point x_1 with $f(x_1) = a$ and a point x_2 with $f(x_2) = b$. Now convince yourself that there is a point on the graph of the function that is also on the line $y = x$.

[138]Exercise 5.8.8. Condition (a) is the intermediate value property (IVP) according to Definition 5.27, while (b) can be interpreted as saying that connectedness is preserved by continuous functions. This latter interpretation requires a careful definition of connectedness in $\mathbb{R}$.

[139]Exercise 5.9.13. You wish to show that (i) f is discontinuous at every point in C, indeed has a jump discontinuity at each such point; (ii) f is continuous at every point not in C; (iii) f is nondecreasing; (iv) f is increasing on any interval in which C is dense; and (v) f is constant on any interval containing no point of C.

The most direct and easiest proof that f is continuous at every point not in C would be to use "uniform convergence" but that is in a later chapter. Here you will have to use an ε-δ argument.

[140]Exercise 5.9.15. How large can the set of discontinuity points be?

[141]Exercise 5.9.16. The function f^{-1} is defined on the interval $J = [f(a), f(b)]$. Explain first why it exists (not all functions must have an inverse). Prove that it is increasing. Prove that it is continuous (using the fact that it is increasing).

[142]Exercise 5.10.1. The equation $f(x+y) = f(x) + f(y)$ is called a functional equation. You are told about this function only that it satisfies such a relationship and has a nice property at one point. Now you must show that this implies more. Show first that $f(0) = 0$ and that $f(x-y) = f(x) - f(y)$.

[143]Exercise 5.10.2. This continues Exercise 5.10.1. Show first that $f(r) = rf(1)$ for all $r = m/n$ rational. Then make use of the continuity of f that you had already established in the other exercise.

[144]Exercise 5.10.3. Show that either f is always zero or else $f(0) = 1$. Establish

$$f(x-y) = f(x)/f(y).$$

[145]Exercise 5.10.5. Consider the intersection

$$\overline{A} \cap \overline{B}.$$

[146]Exercise 5.10.14. You will need to use the fact that

$$\{x : \limsup_{x \to x_0^-} f(x) > \limsup_{x \to x_0^+} f(x)\}$$

is countable. See Exercise 5.10.8.

Chapter 6

MORE ON CONTINUOUS FUNCTIONS AND SETS

✂ This chapter can be considered enrichment material containing also several more advanced topics and may be skipped in its entirety. You can proceed directly to the study of derivatives and integrals with no loss in the continuity of the material.

6.1 Introduction

In this chapter we go much more deeply into the analysis of continuous functions. For this we need some new set theoretic ideas and methods.

6.2 Dense Sets

[This section reviews material from Section 1.9.]

Consider the set $\mathbb{Q}$ of rational numbers and let (a,b) be an open interval in $\mathbb{R}$. How do we show that there is a member of $\mathbb{Q}$ in the interval (a,b); that is, that $(a,b)\cap\mathbb{Q}\neq\emptyset$?

Suppose first that $0<a$. Since $b-a>0$, the archimedean property (Theorem 1.11) implies that there is a positive integer q such that

$$q(b-a)>1.$$

Thus

$$qb>1+qa.$$

The archimedean property also implies that the set of integers

$$\{m\in\mathbb{N}:m>qa\}$$

is nonempty. Thus, according to the well-ordering principle, there is a smallest integer p in this set and for this p, it is true that $p-1\leq qa<p$. It follows that

$$qa<p\leq 1+qa<qb,$$

which implies $a<\frac{p}{q}<b$. We have shown that, under the assumption $a>0$, there exists a rational number $r=p/q$ in the interval (a,b).

The same is true under the assumption $a<0$. To see this observe first that if $a<0<b$, we can take $r=0$. If $a<b<0$, then $0<-b<-a$, so the argument of the

previous paragraph shows that there exists $r \in \mathbb{Q}$ such that $-b < r < -a$. In this case $a < -r < b$.

The preceding discussion proves that every open interval contains a rational number. We often express this fact by saying that the set of rational numbers is a *dense* set.

Definition 6.1: A set of real numbers A is said to be *dense* (in $\mathbb{R}$) if for each open interval (a,b) the set $A \cap (a,b)$ is nonempty.

It is important to have a more general concept, that of a set A being dense in a set B.

Definition 6.2: Let A and B be subsets of $\mathbb{R}$. If every open interval that intersects B also intersects A, we say that A *is dense in* B.

Thus Definition 6.1 states the special case of Definition 6.2 that occurs when $B = \mathbb{R}$. We should note that some authors require that $A \subset B$ in their version of Definition 6.2. We find it more convenient not to impose this restriction. Thus, for example, in *our* language $\mathbb{Q}$ is dense in $\mathbb{R} \setminus \mathbb{Q}$.

It is easy to verify that A is dense in B if and only if $\overline{A} \supset B$ (Exercise 6.2.1).

Exercises

6.2.1 Verify that A is dense in B if and only if $\overline{A} \supset B$.

6.2.2 Prove that every set A is dense in its closure $\overline{A}$.

6.2.3 Prove that if A is dense in B and $C \subset B$, then A is dense in C.

6.2.4 Prove that if $A \subset B$ and A is dense in B, then $\overline{A} = \overline{B}$. Is the statement correct without the assumption that $A \subset B$?

6.2.5 Is $\mathbb{R} \setminus \mathbb{Q}$ dense in $\mathbb{Q}$?

6.2.6 The following are several pairs (A,B) of sets. In each case determine whether A is dense in B.

(a) $A = \mathbb{N}, B = \mathbb{N}$

(b) $A = \mathbb{N}, B = \mathbb{Z}$

(c) $A = \mathbb{N}, B = \mathbb{Q}$

(d) $A = \left\{x : x = \frac{m}{2^n},\ m \in \mathbb{Z}, n \in \mathbb{N}\right\}, B = \mathbb{Q}$

6.2.7 Let A and B be subsets of $\mathbb{R}$. Prove that A is dense in B if and only if for every $b \in B$ there exists a sequence $\{a_n\}$ of points from A such that $\lim_{n\to\infty} a_n = b$.

6.2.8 Let B be the set of all irrational numbers. Prove that the set

$$A = \{q + \sqrt{2} : q \in \mathbb{Q}\}$$

is a countable subset of B that is dense in B.

6.2.9 Let $f : \mathbb{R} \to \mathbb{R}$ be a strictly increasing continuous function. Does f map dense sets to dense sets; that is, is it true that

$$f(E) = \{f(x) : x \in E\}$$

is dense if E is dense?

SEE NOTE 147

6.2.10 Prove that every set $B \subset \mathbb{R}$ contains a countable set A that is dense in B.

6.3 Nowhere Dense Sets

We might view a set A that is dense in $\mathbb{R}$ as being somehow large: Inside every interval, no matter how small, we find points of A. There is an opposite extreme to this situation: A set is said to be *nowhere dense*, and hence is in some sense small, if it is not dense in any interval at all. The precise definition of this important concept of smallness follows.

Definition 6.3: The set $A \subset \mathbb{R}$ is said to be *nowhere dense* in $\mathbb{R}$ provided every open interval I contains an open subinterval J such that $A \cap J = \emptyset$.

We can state this another way: A is nowhere dense provided $\overline{A}$ contains no open intervals. (See Exercise 6.3.4.)

Example 6.4: It is easy to construct examples of nowhere dense sets.

1. Any finite set
2. $\mathbb{N}$
3. $\{1/n : n \in \mathbb{N}\}$

Each of these sets is nowhere dense, as you can verify. ◀

Each of the sets in Example 6.4 is countable and hence also small in the sense of cardinality. It is hard to imagine an uncountable set that is nowhere dense but, as we shall see in Section 6.5, such sets do exist.

We establish a simple result showing that any finite union of nowhere dense sets is again nowhere dense. It is not true that a countable union of nowhere dense sets is again nowhere dense. Indeed countable unions of nowhere dense sets will be important in our subsequent study.

Theorem 6.5: *Let $A_1, A_2, \ldots, A_n$ be nowhere dense in $\mathbb{R}$. Then $A_1 \cup \cdots \cup A_n$ is also nowhere dense in $\mathbb{R}$.*

Proof. Let I be any open interval in $\mathbb{R}$. We seek an open interval $J \subset I$ such that $J \cap A_i = \emptyset$ for $i = 1, 2, \ldots, n$.

Since A_1 is nowhere dense, there exists an open interval $I_1 \subset I$ such that $I_1 \cap A_1 = \emptyset$. Now A_2 is also nowhere dense in $\mathbb{R}$, so there exists an open interval $I_2 \subset I_1$ such that $A_2 \cap I_2 = \emptyset$. Proceeding in this way we obtain open intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_n$$

such that for $i = 1, ..., n$, $A_i \cap I_i = \emptyset$. It follows from the fact that $I_n \subset I_i$ for $i = 1, \ldots, n$ that $A_i \cap I_n = \emptyset$ for $i = 1, \ldots, n$. Thus

$$\left(\bigcup_{i=1}^{n} A_i\right) \cap I_n = \bigcup_{i=1}^{n} (A_i \cap I_n) = \bigcup_{i=1}^{n} \emptyset = \emptyset,$$

as was to be proved. ■

Exercises

6.3.1 Give an example of a sequence of nowhere dense sets whose union is not nowhere dense.

SEE NOTE 148

6.3.2 Which of the following statements are true?

(a) Every subset of a nowhere dense set is nowhere dense.

(b) If A is nowhere dense, then so too is $A + c = \{t + c : t \in A\}$ for every number c.

(c) If A is nowhere dense, then so too is $cA = \{ct : t \in A\}$ for every positive number c.

(d) If A is nowhere dense, then so too is A', the set of derived points of A.

(e) A nowhere dense set can have no interior points.

(f) A set that has no interior points must be nowhere dense.

(g) Every point in a nowhere dense set must be isolated.

(h) If every point in a set is isolated, then that set must be nowhere dense.

SEE NOTE 149

6.3.3 If A is nowhere dense, what can you say about $\mathbb{R} \setminus A$? If A is dense, what can you say about $\mathbb{R} \setminus A$?

6.3.4 Prove that a set $A \subset \mathbb{R}$ is nowhere dense if and only if $\overline{A}$ contains no intervals; equivalently, the interior of $\overline{A}$ is empty.

6.3.5 What should the statement "A is nowhere dense in the interval I" mean? Give an example of a set that is nowhere dense in $[0,1]$ but is not nowhere dense in $\mathbb{R}$.

6.3.6 Let A and B be subsets of $\mathbb{R}$. What should the statement "A is nowhere dense in the B" mean? Is $\mathbb{N}$ nowhere dense in $[0,10]$? Is $\mathbb{N}$ nowhere dense in $\mathbb{Z}$? Is $\{4\}$ nowhere dense in $\mathbb{N}$?

6.3.7 Prove that the complement of a dense open subset of $\mathbb{R}$ is nowhere dense in $\mathbb{R}$.

6.3.8 Let $f : \mathbb{R} \to \mathbb{R}$ be a strictly increasing continuous function. Show that f maps nowhere dense sets to nowhere dense sets; that is,

$$f(E) = \{f(x) : x \in E\}$$

is nowhere dense if E is nowhere dense.

6.4 The Baire Category Theorem

✂ *Advanced section. May be omitted.*

In this section we shall establish the Baire category theorem, which gives a sense in which nowhere dense sets can be viewed as "small:" A union of a sequence of nowhere dense sets cannot fill up an interval. If we interpret Cantor's theorem (Theorem 2.4) as asserting that a union of a sequence of finite sets cannot fill up an interval, then we see the Baire category theorem as a far-reaching generalization.

We motivate this important theorem by way of a game idea that is due to Stefan Banach (1892–1945) and Stanislaw Mazur (1905–1981). Although the origins of the theorem are due to René Baire, after whom the theorem is named, the game approach helps us see why the Baire category theorem might be true. This Banach-Mazur game is just one of many mathematical games that are used throughout mathematics to develop interesting concepts.

6.4.1 A Two-Player Game

✂ *Advanced section. May be omitted.*

We introduce the Baire category theorem via a game between two players (A) and (B).

Player (A) is given a subset A of $\mathbb{R}$, and player (B) is given the complementary set $B = \mathbb{R} \setminus A$. Player (A) first selects a closed interval $I_1 \subset \mathbb{R}$; then player (B) chooses a closed interval $I_2 \subset I_1$. The players alternate moves, a move consisting of selecting a closed interval inside the previously chosen interval.

The play of the game thus determines a descending sequence of closed intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_n \supset \ldots$$

where player (A) chooses those with odd index and player (B) those with even index. If

$$A \cap \bigcap_{n=1}^{\infty} I_n \neq \emptyset,$$

then player (A) wins; otherwise player (B) wins.

The goal of player (A) is evidently to make sure that the intersection contains a point of A; the goal of player (B) is to ensure that the intersection is empty or contains only points of B. We expect that player (A) should win if his set A is large while player (B) should win if his set is large. It is not, however, immediately clear what "large" might mean for this game.

Example 6.6: If the set A given to player (A) contains an open interval J, then (A) should choose any interval $I_1 \subset J$. No matter how the game continues, player (A) wins. Another way to say this: If the set given to player (B) is not dense, he loses. ◀

Example 6.7: For a more interesting example, let player (A) be dealt the "large" set of all irrational numbers, so that player (B) is dealt the rationals. (Both players have been dealt dense sets now.) Let A consist of the irrational numbers. Player (A) can win by following the strategy we now describe. Let $q_1, q_2, q_3, \ldots$ be a listing of all of the

rational numbers; that is,

$$\mathbb{Q} = \{q_1, q_2, q_3, \dots\}.$$

Player (A) chooses the first interval I_1 as any closed interval such that $q_1 \notin I_1$. Inductively, suppose $I_1, I_2, \dots, I_{2n}$ have been chosen according to the rules of the game so that it is now time for player (A) to choose I_{2n+1}. The set $\{q_1, q_2, \dots, q_n\}$ is finite, so there exists a closed interval $I_{2n+1} \subset I_{2n}$ such that

$$I_{2n+1} \cap \{q_1, q_2, \dots, q_n\}$$

is empty. Player (A) chooses such an interval.

Since for each $n \in \mathbb{N}$, $q_n \notin I_{2n+1}$, the set $\bigcap_{n=1}^{\infty} I_n$ contains no rational numbers, but, as a descending sequence of closed intervals, $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$. Thus $A \cap \bigcap_{n=1}^{\infty} I_n \neq \emptyset$, and (A) wins. ◀

In these two examples, using informal language, we can say that player (A) has a *strategy* to win: No matter how player (B) proceeds, player (A) can "answer" each move and win the game.

In both examples player (A) had a clear advantage: The set A was larger than the set B. But in what sense is it larger? It is not the fact that A is uncountable while B is countable that matters here. It is something else: The fact that given an interval I_{2n}, player (A) can choose I_{2n+1} inside I_{2n} in such a way that I_{2n+1} misses the set $\{q_1, q_2, \dots, q_n\}$.

Let us try to see in the second example a general strategy that should work for player (A) in some cases. The set B was the union of the singleton sets $\{q_n\}$. Suppose instead that B is the union of a sequence of "small" sets Q_n. Then the same "strategy" will prevail if given any interval J and given any $n \in \mathbb{N}$, there exists an interval $I \subset J$ such that

$$I \cap (Q_1 \cup Q_2 \cup \cdots \cup Q_n) = \emptyset.$$

The set $\bigcap_{n=1}^{\infty} I_n$ will be nonempty, and will miss the set $\bigcup_{n=1}^{\infty} Q_n$. Thus, if $B = \bigcup_{n=1}^{\infty} Q_n$, player (A) has a winning strategy. It is in this sense that the set B is "small." The set A is "large" because the set B is "small". If we look carefully at the requirement on the sets Q_k, we see it is just that each of these sets is nowhere dense in $\mathbb{R}$.

Thus the key to player (A) winning rests on the concept of a nowhere dense set. But note that it rests on the set B being the union of a sequence of nowhere dense sets.

6.4.2 The Baire Category Theorem

✂ *Advanced section. May be omitted.*

We can formulate our result from our discussion of the game in several ways:

1. $\mathbb{R}$ cannot be expressed as a countable union of nowhere dense sets.

2. The complement of a countable union of nowhere dense sets is dense.

The second of these provides a sense in which countable unions of nowhere dense sets are "small:" No matter which countable collection of nowhere dense sets we choose, their union leaves a dense set uncovered.

To formulate the Baire category theorem we need some definitions. This is the original language of Baire and it has survived; he simply places sets in two types or categories. Into the first category he places the sets that are to be considered small and into the second category he puts the remaining (not small) sets.

Definition 6.8: Let A be a set of real numbers.

1. A is said to be of the *first category* if it can be expressed as a countable union of nowhere dense sets.

2. A is said to be of the *second category* if it is not of the first category.

3. A is said to be *residual* in $\mathbb{R}$ if the complement $\mathbb{R} \setminus A$ is of the first category.

The following properties of first category sets and their complements, the residual sets, are easily proved and left as exercises.

Lemma 6.9: *A union of any sequence of first category sets is again a first category set.*

Lemma 6.10: *An intersection of any sequence of residual sets is again a residual set.*

Theorem 6.11 (Baire Category Theorem) *Every residual subset of $\mathbb{R}$ is dense in $\mathbb{R}$.*

Proof. The discussion in Section 6.4.1 constitutes a proof. Suppose that player (A) is dealt a set $A = X \cap [a,b]$ where X is residual. Then there is a sequence of nowhere dense sets $\{Q_n\}$ so that

$$X = \mathbb{R} \setminus \bigcup_{n=1}^{\infty} Q_n.$$

Then player (A) wins by choosing any interval $I_1 \subset [a,b]$ that avoids Q_1 and continues following the strategy of Section 6.4.1. In particular, X must contain a point of the interval $[a,b]$, and hence a point of any interval. ■

Theorem 6.11 provides a sense of largeness of sets that is not shared by dense sets in general. The intersection of two dense sets might be empty, but the intersection of two, or even countably many, residual sets must still be dense.

Exercises

6.4.1 Show that the union of any sequence of first category sets is again a first category set.
SEE NOTE 150

6.4.2 Show that the intersection of any sequence of residual sets is again a residual set.
SEE NOTE 151

6.4.3 Rewrite the proof of Theorem 6.11 without using the games language.
SEE NOTE 152

6.4.4 Give an example of two dense sets whose intersection is not dense. Does this contradict Theorem 6.11?
SEE NOTE 153

6.4.5 Suppose that $\bigcup_{n=1}^{\infty} A_n$ contains some interval (c,d). Show that there is a set, say A_{n_0}, and a subinterval $(c',d') \subset (c,d)$ so that A_{n_0} is dense in (c',d').
SEE NOTE 154

6.4.3 Uniform Boundedness

✂ *Advanced section. May be omitted.*

There are many applications of the Baire category Theorem in analysis. For now, we present just one application, dealing with the concept of *uniform boundedness*. Suppose we have a collection $\mathcal{F}$ of functions defined on $\mathbb{R}$ with the property that for each $x \in \mathbb{R}$, $\{|f(x)| : f \in \mathcal{F}\}$ is bounded. This means that for each $x \in \mathbb{R}$ there exists a number $M_x \geq 0$ such that $|f(x)| \leq M_x$ for all $f \in \mathcal{F}$. We can describe this situation by saying that $\mathcal{F}$ is *pointwise bounded*. Does this imply that the collection is *uniformly bounded*; that is, that there is a single number M so that $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and every $x \in \mathbb{R}$?

Example 6.12: Let $q_1, q_2, q_3, \ldots$ be an enumeration of $\mathbb{Q}$. For each $n \in \mathbb{N}$ we define a function f_n by $f_n(q_k) = k$ if $n \leq k$, $f_n(x) = 0$ for all other values x. Let $\mathcal{F} = \{f_n : n \in \mathbb{N}\}$. Then if $x \in \mathbb{R} \setminus \mathbb{Q}$, $f(x) = 0$ for all $f \in \mathcal{F}$, and if $x = q_k$, $|f(x)| \leq k$ for all $f \in \mathcal{F}$. Thus, for each $x \in \mathbb{R}$, the set $\{|f(x)| : f \in \mathcal{F}\}$ is bounded. The bounds can be taken to be 0 if $x \in \mathbb{R} \setminus \mathbb{Q}$ ($M_x = 0$ if $x \in \mathbb{R} \setminus \mathbb{Q}$) and we can take $M_{q_k} = k$. But since $\mathbb{Q}$ is dense in $\mathbb{R}$, none of the functions f_n is bounded on any interval. (Verify this.) Thus a collection of functions may be pointwise bounded but not uniformly bounded on any interval. ◀

The functions f_n in Example 6.12 are everywhere discontinuous. Our next theorem shows that if we had taken a collection $\mathcal{F}$ of *continuous* functions, then not only would each $f \in \mathcal{F}$ be bounded on closed intervals (as Theorem 5.49 guarantees), but there would be an interval I on which the entire collection is *uniformly bounded* ; that is, there exists a constant M such that $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and each $x \in I$.

Theorem 6.13: *Let $\mathcal{F}$ be a collection of continuous functions on $\mathbb{R}$ such that for each $x \in \mathbb{R}$ there exists a constant $M_x > 0$ such that $|f(x)| \leq M_x$ for each $f \in \mathcal{F}$. Then there exists an open interval I and a constant $M > 0$ such that $|f(x)| \leq M$ for each $f \in \mathcal{F}$ and $x \in I$.*

Proof. For each $n \in \mathbb{N}$, let $A_n = \{x : |f(x)| \leq n \text{ for all } f \in \mathcal{F}\}$. By hypothesis, $\mathbb{R} = \bigcup_{n=1}^{\infty} A_n$. Also, by hypothesis, each $f \in \mathcal{F}$ is continuous and so it is easy to check that each of the sets

$$\{x : |f(x)| \leq n\}$$

must be closed (e.g., Exercise 5.4.31). Thus

$$A_n = \bigcap_{f \in \mathcal{F}} \{x : |f(x)| \leq n\}$$

is an intersection of closed sets and is therefore itself closed. This expresses the real line $\mathbb{R}$ as a union of the sequence of closed sets $\{A_n\}$.

It now follows from the Baire category theorem that at least one of the sets, say A_{n_0}, must be dense in some open interval I. Since A_{n_0} is closed and dense in the interval I, A_{n_0} must contain I. This means that $|f(x)| \leq n_0$ for each $f \in \mathcal{F}$ and all $x \in I$. ■

Exercises

6.4.6 Let $\{f_n\}$ be a sequence of continuous functions on an interval $[a,b]$ such that

$$\lim_{n\to\infty} f_n(x) = f(x)$$

exists at every point $x \in [a,b]$. Show that f need not be continuous nor even bounded, but that f must be bounded on some subinterval of $[a,b]$.

6.4.7 Let $\{f_n\}$ be a sequence of continuous functions on $[0,1]$ and suppose that

$$\lim_{n\to\infty} f_n(x) = 0$$

for all $0 \leq x \leq 1$. Show that there must be an interval $[c,d] \subset [0,1]$ so that, for all sufficiently large n, $|f_n(x)| \leq 1$ for all $x \in [c,d]$.

SEE NOTE 155

6.4.8 Give an example of a sequence of functions on $[0,1]$ with the property that

$$\lim_{n\to\infty} f_n(x) = 0$$

for all $0 \leq x \leq 1$ and yet for every interval $[c,d] \subset [0,1]$ and every N there is some $x \in [c,d]$ and $n > N$ with $f_n(x) > 1$.

6.5 Cantor Sets

✂ *Advanced section. May be omitted.*

We say that a set is *perfect* if it is a nonempty closed set with no isolated points. The only examples that might come to mind are sets that are finite unions of intervals. It might be difficult to imagine a perfect subset of $\mathbb{R}$ that is also nowhere dense. In this section we obtain such a set, the very important classical Cantor set. We also discuss some of its variants. Such sets have historical significance and are of importance in a number of areas of mathematical analysis.

6.5.1 Construction of the Cantor Ternary Set

✂ *Advanced section. May be omitted.*

We begin with the closed interval $[0,1]$. From this interval we shall remove a dense open set G. The remaining set $K = [0,1] \setminus G$ will then be closed and nowhere dense in [0,1]. We construct G in such a way that K has no isolated points and is nonempty. Thus K will be a nonempty, nowhere dense perfect subset of [0,1].

It is easiest to understand the set G if we construct it in stages. Let $G_1 = \left(\frac{1}{3}, \frac{2}{3}\right)$, and let $K_1 = [0,1] \setminus G_1$. Thus $K_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$ is what remains when the middle third of the interval [0,1] is removed. This is the first stage of our construction.

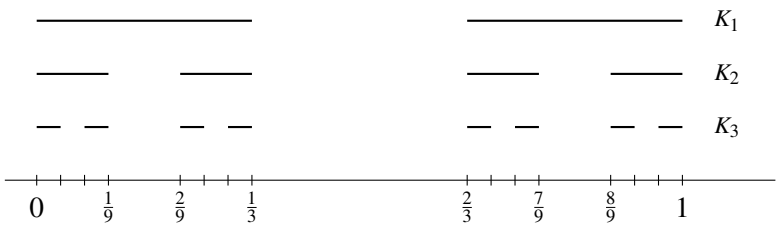


Figure 6.1. The third stage in the construction of the Cantor ternary set.

We repeat this construction on each of the two component intervals of K_1. Let $G_2 = \left(\frac{1}{9}, \frac{2}{9}\right) \cup \left(\frac{7}{9}, \frac{8}{9}\right)$ and let $K_2 = [0,1] \setminus (G_1 \cup G_2)$. Thus

$$K_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$

This completes the second stage.

We continue inductively, obtaining two sequences of sets, $\{K_n\}$ and $\{G_n\}$ with the following properties: For each $n \in \mathbb{N}$

1. G_n is a union of 2^{n-1} pairwise disjoint open intervals.
2. K_n is a union of 2^n pairwise disjoint closed intervals.
3. $K_n = [0,1] \setminus (G_1 \cup G_2 \cup \cdots \cup G_n)$.
4. Each component of G_{n+1} is the "middle third" of some component of K_n.
5. The length of each component of K_n is $1/3^n$.

Figure 6.1 shows K_1, K_2, and K_3.

Now let

$$G = \bigcup_{n=1}^{\infty} G_n$$

and let

$$K = [0,1] \setminus G = \bigcap_{n=1}^{\infty} K_n.$$

Then G is open and the set K (our Cantor set) is closed.

To see that K is nowhere dense, it is enough, since K is closed, to show that K contains no open intervals (Exercise 6.3.4). Let J be an open interval in $[0,1]$ and let λ be its length. Choose $n \in \mathbb{N}$ such that $1/3^n < \lambda$. By property 5, each component of K_n has length $1/3^n < \lambda$, and by property 2 the components of K_n are pairwise disjoint. Thus K_n cannot contain J, so neither can $K = \bigcap_1^\infty K_n$. We have shown that the closed set K contains no intervals and is therefore nowhere dense.

It remains to show that K has no isolated points. Let $x_0 \in K$. We show that x_0 is a limit point of K. To do this we show that for every $\varepsilon > 0$ there exists $x_1 \in K$ such

that $0 < |x_1 - x_0| < \varepsilon$. Choose n such that $1/3^n < \varepsilon$. There is a component L of K_n that contains x_0. This component is a closed interval of length $1/3^n < \varepsilon$. The set $K_{n+1} \cap L$ has two components L_0 and L_1, each of which contains points of K. The point x_0 is in one of the components, say L_0. Let x_1 be any point of $K \cap L_1$. Then $0 < |x_0 - x_1| < \varepsilon$. This verifies that x_0 is a limit point of K. Thus K has no isolated points.

The set K is called the *Cantor set*. Because of its construction, it is often called the Cantor middle third set. In a moment we shall present a purely arithmetic description of the Cantor set that suggests another common name for K, the "Cantor ternary set". But first, we mention a few properties of K and of its complement G that may help you visualize these sets.

First note that G is an open dense set in $[0,1]$. Write $G = \bigcup_{k=1}^{\infty}(a_k, b_k)$. (The component intervals (a_k, b_k) of G can be called the intervals *complementary* to K in $(0,1)$. Each is a middle third of a component interval of some K_n.) Observe that no two of these component intervals can have a common endpoint. If, for example, $b_m = a_n$, then this point would be an isolated point of K, and K has no isolated points.

Next observe that for each $k \in \mathbb{N}$, the points a_k and b_k are points of K. But there are other points of K as well. In fact, we shall see presently that K is uncountable. These other points are all limit points of the endpoints of the complementary intervals. The set of endpoints is countable, but the closure of this set is uncountable as we shall see. Thus, in the sense of cardinality, "most" points of the Cantor set are *not* endpoints of intervals complementary to K.

Each component interval of the set G_n has length $1/3^n$; thus the sum of the lengths of these component intervals is

$$\frac{2^{n-1}}{3^n} = \frac{1}{2}\left(\frac{2}{3}\right)^n.$$

It follows that the lengths of all component intervals of G forms a geometric series with sum

$$\sum_{n=1}^{\infty} \frac{1}{2}\left(\frac{2}{3}\right)^n = 1.$$

(This also gives us a clue as to why K cannot contain an interval: After removing from the unit interval a sequence of pairwise disjoint intervals with length-sum one, no room exists for any intervals in the set K that remains.)

Exercises

6.5.1 Let E be the set of endpoints of intervals complementary to the Cantor set K. Prove that $\overline{E} = K$.

6.5.2 Let G be a dense open subset of $\mathbb{R}$ and let $\{(a_k, b_k)\}$ be its set of component intervals. Prove that $H = \mathbb{R} \setminus G$ is perfect if and only if no two of these intervals have common endpoints.

6.5.3 Let K be the Cantor set and let $\{(a_k, b_k)\}$ be the sequence of intervals complementary to K in $[0,1]$. For each $k \in \mathbb{N}$, let $c_k = (a_k + b_k)/2$ (the midpoint of the interval (a_k, b_k)) and let $N = \{c_k : k \in \mathbb{N}\}$. Prove each of the following:

(a) Every point of N is isolated.

(b) If $c_i \neq c_j$, there exists $k \in \mathbb{N}$ such that c_k is between c_i and c_j (i.e., no point in N has an immediate "neighbor" in N).

(c) Show that there is an *order-preserving mapping* $\phi : \mathbb{Q} \cap (0,1) \to N$ [i.e., if $x < y \in \mathbb{Q} \cap (0,1)$, then $\phi(x) < \phi(y) \in N$]. This may seem surprising since $\mathbb{Q} \cap (0,1)$ has *no* isolated points while N has *only* isolated points.

6.5.4 It is common now to say that a set E of real numbers is a *Cantor set* if it is nonempty, bounded, perfect, and nowhere dense. Show that the union of a finite number of Cantor sets is also a Cantor set.

6.5.5 Show that every Cantor set is uncountable.

6.5.6 ✂ Let A and B be subsets of $\mathbb{R}$. A function h that maps A onto B, is one-to-one, and with both h and h^{-1} continuous is called a *homeomorphism* between A and B. The sets A and B are said to be *homeomorphic*. Prove that a set C is a Cantor set if and only if it is homeomorphic to the Cantor ternary set K.

6.5.2 An Arithmetic Construction of K

✂ *Enrichment section. May be omitted.*

We turn now to a purely arithmetical construction for the Cantor set. You will need some familiarity with ternary (base 3) arithmetic here.

Each $x \in [0,1]$ can be expressed in base 3 as

$$x = .a_1a_2a_3\ldots,$$

where $a_i = 0$, 1 or 2, $i = 1,2,3,\ldots$. Certain points have two representations, one ending with a string of zeros, the other in a string of twos. For example, $.1000\cdots = .0222\ldots$ both represent the number 1/3 (base ten). Now, if $x \in (1/3, 2/3)$, $a_1 = 1$, thus each $x \in G_1$ must have '1' in the first position of its ternary expansion. Similarly, if

$$x \in G_2 = \left(\frac{1}{9}, \frac{2}{9}\right) \cup \left(\frac{7}{9}, \frac{8}{9}\right),$$

it must have a 1 in the second position of its ternary expansion (i.e., $a_2 = 1$). In general, each point in G_n must have $a_n = 1$. It follows that every point of $G = \bigcup_1^\infty G_n$ must have a 1 someplace in its ternary expansion.

Now endpoints of intervals complementary to K have two representations, one of which involves no 1's. The remaining points of K never fall in the middle third of a component of one of the sets K_n, and so have ternary expansions of the form

$$x = .a_1a_2\ldots \quad a_i = 0 \text{ or } 2.$$

We can therefore describe K arithmetically as the set

$$\{x = .a_1a_2a_3\ldots \text{ (base three) } : a_i = 0 \text{ or } 2 \text{ for each } i \in \mathbb{N}\}.$$

As an immediate result, we see that K is uncountable. In fact, K can be put into 1-1 correspondence with [0,1]: For each

$$x = .a_1a_2a_3\ldots \text{ (base 3)}, a_i = 0,2,$$

in the set K, let there correspond the number

$$y = .b_1b_2b_3\ldots \text{ (base 2)}, b_i = a_i/2.$$

This provides a 1-1 correspondence between K (minus endpoints of complementary intervals) and $[0,1]$ (minus the countable set of numbers with two base 2 representations). By allowing these two countable sets to correspond to each other, we obtain a 1-1 correspondence between K and $[0,1]$.

Note. We end this section by mentioning that variations in the constructions of K can lead to interesting situations. For example, by changing the construction slightly, we can remove intervals in such a way that

$$G' = \bigcup_{k=1}^{\infty}(a'_k, b'_k)$$

with

$$\sum_{k=1}^{\infty}(b'_k - a'_k) = 1/2$$

(instead of 1), while still keeping $K' = [0,1] \setminus G'$ nowhere dense and perfect. The resulting set K' created problems for late nineteenth-century mathematicians trying to develop a theory of measure. The "measure" of G' should be 1/2; the "measure" of [0,1] should be 1. Intuition requires that the measure of the nowhere dense set K' should be $1 - \frac{1}{2} = \frac{1}{2}$. How can this be when K' is so "small?"

Exercises

6.5.7 Find a specific irrational number in the Cantor ternary set.

SEE NOTE 156

6.5.8 Show that the Cantor ternary set can be defined as

$$K = \left\{ x \in [0,1] : x = \sum_{n=1}^{\infty} \frac{i_n}{3^n} \text{ for } i_n = 0 \text{ or } 2 \right\}.$$

6.5.9 Let

$$D = \left\{ x \in [0,1] : x = \sum_{n=1}^{\infty} \frac{j_n}{3^n} \text{ for } j_n = 0 \text{ or } 1 \right\}.$$

Show that $D + D = \{x + y : x, y \in D\} = [0,1]$. From this deduce, for the Cantor ternary set K, that $K + K = [0,2]$.

6.5.10 A careless student makes the following argument. Explain the error.

> "If $G = (a,b)$, then $\overline{G} = [a,b]$. Similarly, if $G = \bigcup_{i=1}^{\infty}(a_i,b_i)$ is an open set, then $\overline{G} = \bigcup_{i=1}^{\infty}[a_i,b_i]$. It follows that an open set G and its closure $\overline{G}$ differ by at most a countable set."

SEE NOTE 157

6.5.3 The Cantor Function

✂ *Advanced section. May be omitted.*

The Cantor set allows the construction of a rather bizarre function that is continuous and nondecreasing on the interval $[0,1]$. It has the property that it is constant on every interval complementary to the Cantor set and yet manages to increase from $f(0)=0$ to $f(1)=1$ by doing all of its increasing on the Cantor set itself. It has sometimes been called "the devil's staircase."

Define the function f in the following way. On $(1/3,2/3)$, let $f=1/2$; on $(1/9,2/9)$, let $f=1/4$; on $(7/9,8/9)$, let $f=3/4$. Proceed inductively. On the 2^{n-1} open intervals appearing at the nth stage, define f to satisfy the following conditions:

1. f is constant on each of these intervals.
2. f takes the values
$$\frac{1}{2^n}, \frac{3}{2^n}, \ldots, \frac{2^n-1}{2^n}$$
on these intervals.
3. If x and y are members of different nth-stage intervals with $x<y$, then $f(x)<f(y)$.

This description defines f on $G=[0,1]\setminus K$. Extend f to all of $[0,1]$ by defining $f(0)=0$ and, for $0<x\leq 1$,
$$f(x)=\sup\{f(t):t\in G, t<x\}.$$
In order to check that this defines the function that we want, we need to check each of the following.

1. $f(G)$ is dense in $[0,1]$.
2. f is nondecreasing on $[0,1]$.
3. f is continuous on $[0,1]$.
4. $f(K)=[0,1]$.

These have been left as exercises.

Figure 6.2 illustrates the construction. The function f is called the *Cantor function*. Observe that f "does all its rising" on the set K.

The Cantor function allows a negative answer to many questions that might be asked about functions and derivatives and, hence, has become a popular counterexample. For example, let us follow this kind of reasoning. If f is a continuous function on $[0,1]$ and $f'(x)=0$ for every $x\in(0,1)$ then f is constant. (This is proved in most calculus courses by using the mean value theorem.) Now suppose that we know less, that $f'(x)=0$ for every $x\in(0,1)$ excepting a "small" set E of points at which we know nothing. If E is finite it is still easy to show that f must be constant. If E is countable it is possible, but a bit more difficult, to show that it is still true that f must be constant. The question then arises, just how small a set E can appear here; that is, what would we have to know

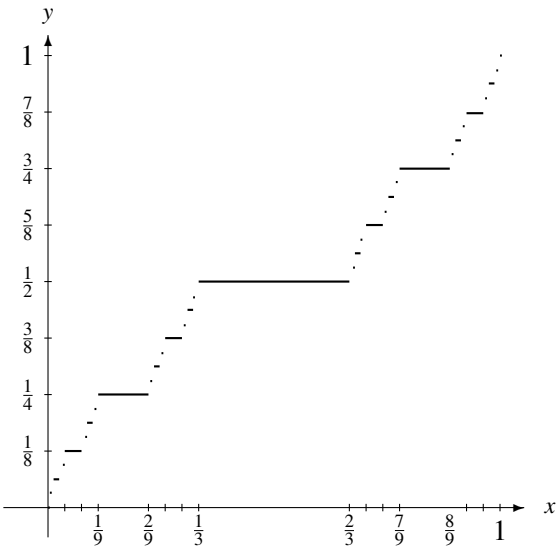


Figure 6.2. The third stage in the construction of the Cantor function.

about a set E so that we could say $f'(x) = 0$ for every $x \in (0,1) \setminus E$ implies that f is constant?

The Cantor function is an example of a function constant on every interval complementary to the Cantor set K (and so with a zero derivative at those points) and yet is not constant. The Cantor set, since it is nowhere dense, might be viewed as extremely small, but even so it is not insignificant for this problem.

Exercises

6.5.11 In the construction of the Cantor function complete the verification of details.

(a) Show that $f(G)$ is dense in $[0,1]$.

(b) Show that f is nondecreasing on $[0,1]$.

(c) Infer from (a) and (b) that f is continuous on $[0,1]$.

(d) Show that $f(K) = [0,1]$ and thus (again) conclude that K is uncountable.

6.5.12 Find a calculus textbook proof for the statement that a continuous function f on an interval $[a,b]$ that has a zero derivative on (a,b) must be constant. Improve the proof to allow a finite set of points on which f is not known to have a zero derivative.

6.6 Borel Sets

✂ *Advanced section. May be omitted.*

In our study of continuous functions we have seen that the classes of open sets and closed sets play a significant role. But the class of sets that are of importance in analysis goes beyond merely the open and closed sets. E. Borel (1871–1956) recognized that for many operations of analysis we need to form countable intersections and countable

unions of classes of sets. The collection of Borel sets was introduced exactly to allow these operations. We recall that a countable union of closed sets may not be closed (or open) and that a countable intersection of open sets, also, may not be open (or closed).

In this section we introduce two additional types of sets of importance in analysis, sets of type $\mathcal{G}_\delta$ and sets of type $\mathcal{F}_\sigma$. These classes form just the beginning of the large class of Borel sets. We shall find that they are precisely the right classes of sets to solve some fundamental questions about real functions.

6.6.1 Sets of Type G_δ

✂ *Advanced section. May be omitted.*

Recall that the union of a collection of open sets is open (regardless of how many sets are in the collection), but the intersection of a collection of open sets need not be open if the collection has infinitely many sets. For example,

$$\bigcap_{n=1}^{\infty}\left(-\frac{1}{n},\frac{1}{n}\right)=\{0\}.$$

Similarly, if $q_1,q_2,q_3,\ldots$ is an enumeration of $\mathbb{Q}$, then

$$\bigcap_{k=1}^{\infty}(\mathbb{R}\setminus\{q_k\})=\mathbb{R}\setminus\mathbb{Q},$$

the set of irrational numbers. The set $\{0\}$ is closed (not open), and $\mathbb{R}\setminus\mathbb{Q}$ is neither open nor closed. The set $\mathbb{R}\setminus\mathbb{Q}$ is a countable intersection of open sets. Such sets are of sufficient importance to give them a name.

Definition 6.14: A subset H of $\mathbb{R}$ is said to be of *type* $\mathcal{G}_\delta$ (or a $\mathcal{G}_\delta$ *set*) if it can be expressed as a countable intersection of open sets, that is, if there exist open sets $G_1,G_2,G_3,\ldots$ such that $H=\bigcap_{k=1}^{\infty}G_k$.

Example 6.15: A closed interval $[a,b]$ or a half-open interval $(a,b]$ is of type $\mathcal{G}_\delta$ since

$$[a,b]=\bigcap_{n=1}^{\infty}\left(a-\frac{1}{n},b+\frac{1}{n}\right)$$

and

$$(a,b]=\bigcap_{n=1}^{\infty}\left(a,b+\frac{1}{n}\right).$$

◀

Theorem 6.16: *Every open set and every closed set in* $\mathbb{R}$ *is of type* $\mathcal{G}_\delta$.

Proof. Let G be an open set in $\mathbb{R}$. It is clear that G is of type $\mathcal{G}_\delta$. We also show that G can be expressed as a countable union of closed sets. Express G in the form

$$G=\bigcup_{k=1}^{\infty}(a_k,b_k)$$

where the intervals (a_k, b_k) are pairwise disjoint. Now for each $k \in \mathbb{N}$ there exist sequences $\{c_{k_j}\}$ and $\{d_{k_j}\}$ such that the sequence $\{c_{k_j}\}$ decreases to a_k, the sequence $\{d_{k_j}\}$ increases to b_k and $c_{k_j} < d_{k_j}$ for each $j \in \mathbb{N}$. Thus

$$(a_k, b_k) = \bigcup_{j=1}^{\infty} [c_{k_j}, d_{k_j}].$$

We have expressed each component interval of G as a countable union of closed sets. It follows that

$$G = \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{\infty} [c_{k_j}, d_{k_j}] = \bigcup_{j,k=1}^{\infty} [c_{k_j}, d_{k_j}]$$

is also a countable union of closed sets. Now take complements. This shows that $\mathbb{R} \setminus G$ can be expressed as a countable intersection of open sets (by using the de Morgan laws). Since every closed set F can be written

$$F = \mathbb{R} \setminus G$$

for some open set G, we have shown that any closed set is of type $\mathcal{G}_\delta$. ■

We observed in Section 6.4 that a dense set can be small in the sense of category. For example, $\mathbb{Q}$ is a first category set. Our next result shows that a dense set *of type* $\mathcal{G}_\delta$ must be large in the sense of category.

Theorem 6.17: *Let H be of type $\mathcal{G}_\delta$ and be dense in $\mathbb{R}$. Then H is residual.*

Proof. Write

$$H = \bigcap_{k=1}^{\infty} G_k$$

with each of the sets G_k open. Since H is dense by hypothesis and $H \subset G_k$ for each $k \in \mathbb{N}$, each of the open sets G_k is also dense. Thus $\mathbb{R} \setminus G_k$ is nowhere dense for every $k \in \mathbb{N}$, and so each G_k is residual. The result now follows from Lemma 6.10. ■

Exercises

6.6.1 Which of the following sets are of type $\mathcal{G}_\delta$?

(a) $\mathbb{N}$

(b) $\left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$

(c) The set $\{C_n : n \in \mathbb{N}\}$ of midpoints of intervals complementary to the Cantor set

(d) A finite union of intervals (that need not be open or closed)

6.6.2 Prove Theorem 6.17 for the interval $[a, b]$ in place of $\mathbb{R}$.

6.6.3 Prove that a set E of type $\mathcal{G}_\delta$ in $\mathbb{R}$ is either residual or else there is an interval containing no points of E.

6.6.2 Sets of Type F_σ

✂ *Advanced section. May be omitted.*

Just as the countable intersections of open sets form a larger class of sets, the $\mathcal{G}_\delta$ sets, so also the countable unions of closed sets form a larger class of sets.

The complements of open sets are closed. By dealing with complements of $\mathcal{G}_\delta$ sets we arrive at the dual notion of a set of type $\mathcal{F}_\sigma$.

Definition 6.18: A subset E of $\mathbb{R}$ is said to be of *type* $\mathcal{F}_\sigma$ (or an $\mathcal{F}_\sigma$ set) if it can be expressed as a countable union of closed sets; that is, if there exist closed sets $F_1, F_2, F_3, \ldots$ such that $E = \bigcup_{k=1}^{\infty} F_k$.

Using the de Morgan laws, we verify easily that the complement of a $\mathcal{G}_\delta$ set is an $\mathcal{F}_\sigma$ and vice versa (Exercise 6.6.4). This is closely related to the fact that a set is open if and only if its complement is closed.

Example 6.19: The set of rational numbers, $\mathbb{Q}$ is a set of type $\mathcal{F}_\sigma$. This is clear since it can be expressed as

$$\mathbb{Q} = \bigcup_{n=1}^{\infty} \{r_n\}$$

where $\{r_n\}$ is any enumeration of the rationals. The singleton sets $\{r_n\}$ are clearly closed. But note that $\mathbb{Q}$ is *not* of type $\mathcal{G}_\delta$ also. It follows from Theorem 6.17 that a dense set of type $\mathcal{G}_\delta$ must be uncountable (because a countable set is first category). In particular, $\mathbb{Q}$ is not of type $\mathcal{G}_\delta$ (and therefore $\mathbb{R} \setminus \mathbb{Q}$ is not of type $\mathcal{F}_\sigma$). ◀

Theorem 6.20: *A set is of type $\mathcal{G}_\delta$ if and only if its complement is of type $\mathcal{F}_\sigma$.*

Example 6.21: A half-open interval $(a,b]$ is both of type $\mathcal{G}_\delta$ and of type $\mathcal{F}_\sigma$:

$$(a,b] = \bigcap_{n=1}^{\infty} \left(a, b + \frac{1}{n}\right) = \bigcup_{n=1}^{\infty} \left[a + \frac{b-a}{n}, b\right].$$

◀

Note. The only subsets of $\mathbb{R}$ that are both open and closed are the empty set and $\mathbb{R}$ itself. There are, however, many sets that are of type $\mathcal{G}_\delta$ and also of type $\mathcal{F}_\sigma$. See Exercise 6.6.1.

We can now enlarge on Theorem 6.16. There we showed that all open sets and all closed sets are in the class $\mathcal{G}_\delta$. We now show they are also in the class $\mathcal{F}_\sigma$.

Theorem 6.22: *Every open set and every closed set in $\mathbb{R}$ is both of type $\mathcal{F}_\sigma$ and $\mathcal{G}_\delta$.*

Proof. In the proof of Theorem 6.16 we showed explicitly how to express any open set as an $\mathcal{F}_\sigma$. Thus open sets are of type $\mathcal{F}_\sigma$ as well as of type $\mathcal{G}_\delta$ (the latter being trivial). The part pertaining to closed sets now follows by considering complements and using the de Morgan laws. The complement of a closed set is open and therefore the complement of an $\mathcal{F}_\sigma$ set is a $\mathcal{G}_\delta$ set. ■

Exercises

6.6.4 Verify that a subset A of $\mathbb{R}$ is an $\mathcal{F}_\sigma$ ($\mathcal{G}_\delta$) if and only if $\mathbb{R} \setminus A$ is a $\mathcal{G}_\delta$ ($\mathcal{F}_\sigma$).

6.6.5 Which of the following sets are of type $\mathcal{F}_\sigma$?

(a) $\mathbb{N}$

(b) $\left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$

(c) The set $\{C_n : n \in \mathbb{N}\}$ of midpoints of intervals complementary to the Cantor set

(d) A finite union of intervals (that need not be open or closed)

6.6.6 ✂ Prove that a set of type $\mathcal{F}_\sigma$ in $\mathbb{R}$ is either first category or contains an open interval.

6.6.7 ✂ Let $\{f_n\}$ be a sequence of real functions defined on $\mathbb{R}$ and suppose that $f_n(x) \to f(x)$ at every point x. Show that

$$\{x : f(x) > \alpha\} = \bigcup_{m=1}^{\infty} \bigcup_{r=1}^{\infty} \bigcap_{n=r}^{\infty} \{x : f_n(x) \geq \alpha + 1/m\}.$$

If each function f_n is continuous, what can you assert about the set

$$\{x : f(x) > \alpha\}?$$

See Note 158

6.7 Oscillation and Continuity

✂ *Advanced section. May be omitted.*

In this section we return to a problem that we began investigating in Section 5.9 about the nature of the set of discontinuity points of a function. To discuss this set we shall need the notions of $\mathcal{F}_\sigma$ and $\mathcal{G}_\delta$ sets and we need to introduce a new tool, the oscillation of a function.

We begin with an example of a function f that is discontinuous at every rational number and continuous at every irrational number.

Example 6.23: Let $q_1, q_2, q_3, \ldots$ be an enumeration of $\mathbb{Q}$. Define a function f by

$$f(x) = \begin{cases} \frac{1}{k}, & \text{if } x = q_k \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Since $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, f can be continuous at a point x only if $f(x) = 0$; that is, only if $x \in \mathbb{R} \setminus \mathbb{Q}$. Thus f is discontinuous at every $x \in \mathbb{Q}$. To check that f is continuous at each point of $\mathbb{R} \setminus \mathbb{Q}$, let $x_0 \in \mathbb{R} \setminus \mathbb{Q}$ and let $\varepsilon > 0$. Choose $k \in \mathbb{N}$ such that $1/k < \varepsilon$. Since the set $q_1, q_2, \ldots, q_k$ is a finite set not containing x_0, there exists $\delta > 0$ such that $|q_i - x_0| \geq \delta$ for each $i = 1, \ldots, k$. Thus if $x \in \mathbb{R}$ and $|x - x_0| < \delta$, then either $x \in \mathbb{R} \setminus \mathbb{Q}$ or $x = q_j$ for some $j > k$. In either case $|f(x) - f(x_0)| \leq \frac{1}{k} < \varepsilon$. This verifies the continuity of f at x_0. Since x_0 was an arbitrary irrational point, we see that f is continuous at every irrational. ◀

Our example shows that it is possible for a function to be continuous at every irrational number and discontinuous at every rational number. Is it possible for the opposite

to occur? Does there exist a function f continuous on $\mathbb{Q}$ and discontinuous on $\mathbb{R} \setminus \mathbb{Q}$? More generally, what sets can be the set of points of continuity of some function f defined on an interval.

We answer this question in this section. The principal tool is that of *oscillation* of a function at a point.

6.7.1 Oscillation of a Function

✂ *Advanced section. May be omitted.*

In order to describe a point of discontinuity we need a way of measuring that discontinuity. For monotonic functions the jump was used previously for such a measure. For general, nonmonotonic, functions a different tool is used.

Definition 6.24: Let f be defined on a nondegenerate interval I. We define the *oscillation of f on I* as the quantity

$$\omega f(I) = \sup_{x,y \in I} |f(x) - f(y)|.$$

Let's see how oscillation relates to continuity. Suppose f is defined in a neighborhood of x_0, and f is continuous at x_0. Then

$$\inf_{\delta > 0} \omega f((x_0 - \delta, x_0 + \delta)) = 0. \tag{1}$$

To see this, let $\varepsilon > 0$. Since f is continuous at x_0, there exists $\delta_0 > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon/2$$

if $|x - x_0| < \delta_0$. If

$$x_0 - \delta_0 < x_1 \leq x_2 < x_0 + \delta_0,$$

then

$$|f(x_1) - f(x_2)| \leq |f(x_1) - f(x_0)| + |f(x_0) - f(x_2)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \tag{2}$$

Since (2) is valid for all $x_1, x_2 \in (x_0 - \delta_0, x_0 + \delta_0)$, we have

$$\sup\{|f(x_1) - f(x_2)| : x_0 - \delta_0 < x_1 \leq x_2 < x_0 + \delta_0\} \leq \varepsilon. \tag{3}$$

But (3) implies that if $0 < \delta < \delta_0$, then

$$\omega f([x_0 - \delta, x_0 + \delta]) \leq \varepsilon.$$

Since ε was arbitrary, the result follows.

The converse is also valid. Suppose (1) holds. Let $\varepsilon > 0$. Choose $\delta > 0$ such that

$$\omega f(x_0 - \delta, x_0 + \delta) < \varepsilon.$$

Then

$$\sup\{|f(x) - f(x_0)| : x \in (x_0 - \delta, x_0 + \delta)\} < \varepsilon,$$

so $|f(x) - f(x_0)| < \varepsilon$ whenever $|x - x_0| < \delta$. This implies continuity of f at x_0.

We summarize the preceding as a theorem.

Theorem 6.25: *Let f be defined on an interval I and let $x_0 \in I$. Then f is continuous at x_0 if and only if*

$$\inf_{\delta>0} \omega f((x_0-\delta, x_0+\delta)) = 0.$$

The quantity in the statement of the theorem is sufficiently important to have a name.

Definition 6.26: Let f be defined in a neighborhood of x_0. The quantity

$$\omega_f(x_0) = \inf_{\delta>0} \omega f((x_0-\delta, x_0+\delta))$$

is called the *oscillation of f at x_0*.

Theorem 6.25 thus states that a function f is continuous at a point x_0 if and only if $\omega_f(x_0) = 0$. Returning to the function that introduced this section, we see that

$$\omega_f(x) = \begin{cases} 1/k, & \text{if } x = q_k \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Let's now see how the concept of oscillation relates to the set of points of continuity of a function.

Theorem 6.27: *Let f be defined on a closed interval I (which may be all of $\mathbb{R}$). Let $\gamma > 0$. Then the set*

$$\{x : \omega_f(x) < \gamma\}$$

is open and the set

$$\{x : \omega_f(x) \geq \gamma\}$$

is closed.

Proof. Let $A = \{x : \omega_f(x) < \gamma\}$ and let $x_0 \in A$. We wish to find a neighborhood U of x_0 such that $U \subset A$; that is, such that $\omega_f(x) < \gamma$ for all $x \in U$.

Let $\omega_f(x_0) = \alpha < \gamma$ and let $\beta \in (\alpha, \gamma)$. From Definition 6.26 we infer the existence of a number $\delta > 0$ such that

$$|f(u) - f(v)| \leq \beta$$

for $u, v \in (x_0-\delta, x_0+\delta)$. Let

$$U = (x_0-\delta, x_0+\delta)$$

and let $x \in U$. Since U is open, there exists $\delta_1 < \delta$ such that

$$(x-\delta_1, x+\delta_1) \subset U.$$

Then

$$\begin{aligned} \omega_f(x) &\leq \sup\{|f(t)-f(s)| : t,s \in (x-\delta_1, x+\delta_1)\} \\ &\leq \sup\{|f(u)-f(v)| : u,v \in U\} \leq \beta < \gamma, \end{aligned}$$

so $x \in A$. This proves A is open. It follows then that the complement of A in I, the set

$$\left\{x : \omega_f(x) \geq \gamma\right\},$$

must be closed. ■

We use the oscillation in the next subsection to answer a question about the nature of the set of points of continuity of a function.

Exercises

6.7.1 Suppose that f is bounded on an interval I. Prove that

$$\omega f(I) = \sup_{x \in I} f(x) - \inf_{x \in I} f(x).$$

6.7.2 A careless student believes that the oscillation can be written as

$$\omega_f(x_0) = \limsup_{x \to x_0} f(x) - \liminf_{x \to x_0} f(x).$$

Show that this is not true, even for bounded functions.

6.7.3 Prove that

$$\omega_f(x_0) = \lim_{\delta \to 0+} \omega f((x_0 - \delta, x_0 + \delta)).$$

See Note 159

6.7.4 Calculate $\omega_f(0)$ for each of the following functions.

(a) $f(x) = \begin{cases} x, & \text{if } x \neq 0 \\ 4, & \text{if } x = 0 \end{cases}$

(b) $f(x) = \begin{cases} 0, & \text{if } x \in \mathbb{Q} \\ 1, & \text{if } x \notin \mathbb{Q} \end{cases}$

(c) $f(x) = \begin{cases} n, & \text{if } x = \frac{1}{n} \\ 0, & \text{otherwise} \end{cases}$

(d) $f(x) = \begin{cases} \sin\frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

(e) $f(x) = \begin{cases} \sin\frac{1}{x}, & \text{if } x \neq 0 \\ 7, & \text{if } x = 0 \end{cases}$

(f) $f(x) = \begin{cases} \frac{1}{x}\sin\frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

6.7.5 In the proof of Theorem 6.27 we let $\omega_f(x_0) = \alpha < \gamma$ and let $\beta \in (\alpha, \gamma)$. Why was the β introduced? Would the proof have worked if we had used $\beta = \gamma$?

6.7.2 The Set of Continuity Points

✂ *Advanced section. May be omitted.*

Given an arbitrary function, how can we describe the nature of the set of points where f is continuous? Can it be any set? Given a set E, how can we know whether there is a function that is continuous at every point of E and discontinuous at every point not in E?

We saw in Example 6.23 that a function exists whose set of continuity points is exactly the irrationals. Can a function exist whose set of continuity points is exactly the rationals? By characterizing the set of such points we can answer this and other questions about the structure of functions.

We now prove the main result of this section using primarily the notion of oscillation introduced in Section 6.7.1.

Theorem 6.28: *Let f be defined on a closed interval I (which may be all of $\mathbb{R}$). Then the set C_f of points of continuity of f is of type $\mathcal{G}_\delta$, and the set D_f of points of discontinuity of f is of type $\mathcal{F}_\sigma$. Conversely, if H is a set of type $\mathcal{G}_\delta$, then there exists a function f defined on $\mathbb{R}$ such that $C_f = H$.*

Proof. To prove the first part, let $f : I \to \mathbb{R}$. We show that the set

$$C_f = \{x : \omega_f(x) = 0\}$$

is of type $\mathcal{G}_\delta$. For each $k \in \mathbb{N}$, let

$$B_k = \left\{x : \omega_f(x) \geq \frac{1}{k}\right\}.$$

By Theorem 6.27, each of the sets B_k is closed. Thus the set

$$B = \bigcup_{k=1}^{\infty} B_k$$

is of type $\mathcal{F}_\sigma$. By Theorem 6.25, $D_f = B$. Therefore, $C_f = I \setminus B$. Since the complement of an $\mathcal{F}_\sigma$ is a $\mathcal{G}_\delta$, the set C_f is a $\mathcal{G}_\delta$.

To prove the converse, let H be any subset of $\mathbb{R}$ of type $\mathcal{G}_\delta$. Then H can be expressed in the form

$$H = \bigcap_{k=1}^{\infty} G_k$$

with each of the sets G_k being open. We may assume without loss of generality that $G_1 = \mathbb{R}$ and that $G_i \supset G_{i+1}$ for each $i \in \mathbb{N}$. (Verify this.)

Let $\{\alpha_k\}$ and $\{\beta_k\}$ be sequences of positive numbers, each converging to zero, with

$$\alpha_k > \beta_k > \alpha_{k+1},$$

for all $k \in \mathbb{N}$. Define a function $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \in H \\ \alpha_k & \text{if } x \in (G_k \setminus G_{k+1}) \cap \mathbb{Q} \\ \beta_k & \text{if } x \in (G_k \setminus G_{k+1}) \cap (\mathbb{R} \setminus \mathbb{Q}). \end{cases}$$

We show that f is continuous at each point of H and discontinuous at each point of $\mathbb{R} \setminus H$.

Let $x_0 \in H$ and let $\varepsilon > 0$. Choose n such that $\alpha_n < \varepsilon$. Since

$$x_0 \in H = \bigcap_{k=1}^{\infty} G_k,$$

we see that $x_0 \in G_n$. The set G_n is open, so there exists $\delta > 0$ such that $(x_0 - \delta, x_0 + \delta) \subset G_n$. From the definition of f on G_n, we see that

$$0 \leq f(x) \leq \alpha_n < \varepsilon$$

for all $x \in (x_0 - \delta, x_0 + \delta)$. Thus

$$|f(x) - f(x_0)| = |f(x) - 0| = |f(x)| < \varepsilon$$

if $|x - x_0| < \delta$, so f is continuous at x_0.

Now let $x_0 \in \mathbb{R} \setminus H$. Then there exists $k \in \mathbb{N}$ such that x_0 belongs to the set $G_k \setminus G_{k+1}$. Thus $f(x_0) = \alpha_k$ or $f(x_0) = \beta_k$. Let us suppose that $f(x_0) = \alpha_k$. If x_0 is an interior point of $G_k \setminus G_{k+1}$, then x_0 is a limit point of

$$\{x : x \in (G_k \setminus G_{k+1}) \cap (\mathbb{R} \setminus \mathbb{Q})\} = \{x : f(x) = \beta_k\},$$

so f is discontinuous at x_0.

The argument is similar if x_0 is a boundary point of $G_k \setminus G_{k+1}$. Again, assume $f(x_0) = \alpha_k$. Arbitrarily close to x_0 there are points of the set

$$\mathbb{R} \setminus (G_k \setminus G_{k+1}).$$

At these points, f takes on values in the set

$$S = \{0\} \cup \bigcup_{i \neq k} \alpha_i \cup \bigcup_{j \neq k} \beta_j.$$

The only limit point of this set is zero and so S is closed. In particular, α_k is *not* a limit point of this set and does not belong to the set. Let ε be half the distance from the point α_k to the closed set S; that is, let

$$\varepsilon = \frac{1}{2} d(\alpha_k, S).$$

Arbitrarily close to x_0 there are points x such that $f(x) \in S$. For such a point,

$$|f(x) - f(x_0)| = |f(x) - \alpha_k| > \varepsilon,$$

so f is discontinuous at x_0. ■

Observe that Theorem 6.28 answers a question we asked earlier: Is there a function f continuous on $\mathbb{Q}$ and discontinuous at every point of $\mathbb{R} \setminus \mathbb{Q}$? The answer is negative, since $\mathbb{Q}$ is not of type $\mathcal{G}_\delta$.

Exercises

6.7.6 In the second part of the proof of Theorem 6.28 we provided a construction for a function f with $C_f = H$, where H is an arbitrary set of type $\mathcal{G}_\delta$. Exhibit explicitly sets G_k that will give rise to a function f such that $C_f = \mathbb{R} \setminus \mathbb{Q}$. Can you do this in such a way that the resulting function is the one we obtained at the beginning of this section?

6.7.7 In the proof of Theorem 6.28 we took $\varepsilon = \frac{1}{2} d(\alpha_k, S)$. Show that this number equals

$$\frac{1}{2} \min_{i \neq k} \{\min\{|\alpha_i - \alpha_k|, |\beta_i - \beta_k|\}\}.$$

6.8 Sets of Measure Zero

✂ *Advanced section. May be omitted.*

In analysis there are a number of ways in which a set might be considered as "small." For example, the Cantor set is not small in the sense of counting: It is uncountable. It is small in another different sense: It is nowhere dense, that is there is no interval at all in which it is dense. Now we turn to another way in which the Cantor set can be considered small: It has "zero length."

Example 6.29: Suppose we wish to measure the "length" of the Cantor set. Since the Cantor set is rather bizarre, we might look instead at the sequence of intervals that have been removed. There is no difficulty in assigning a meaning of length to an interval; the length of (a,b) is $b-a$. What is the total length of the intervals removed in the construction of the Cantor set? From the interval $[0,1]$ we remove first a middle third interval of length $1/3$, then two middle third intervals of length $1/9$, and so on so that at the nth stage we remove 2^{n-1} intervals each of length 3^{-n}. The sum of the lengths of all intervals so removed is

$$1/3+2(1/9)+4(1/27)+\cdots=$$

$$1/3\,(1+2/3+(2/3)^2+(2/3)^3+\ldots)=1.$$

From the interval $[0,1]$ we appear to have removed all of the length. What is left over, the Cantor set, must have length zero.

This method of computing lengths has some merit but it is not the one we wish to adopt here. Another approach to "measuring" the length of the Cantor set is to consider the length that *remains* at each stage. At the first stage the Cantor set is contained inside the union

$$[0,1/3]\cup[2/3,1],$$

which has length $2(1/3)$. At the next stage it is contained inside a union of four intervals, with total length $4(1/9)$. Similarly, at the nth stage the Cantor set is contained inside the union of 2^n intervals each of length 3^{-n}. The sum of the lengths of all these intervals is $(2/3)^n$, and this tends to zero as n gets large. Thus, as before, it seems we should assign zero length to the Cantor set. ◀

We convert the second method of the example into a definition of what it means for a set to be of measure zero. "Measure" is the technical term used to describe the "length" of sets that need not be intervals. In the example we used closed intervals while in our definition we have employed open intervals. There is no difference (see Exercise 6.8.13). In the example we covered the Cantor set with a finite sequence of intervals while in our definition we have employed an infinite sequence. For the Cantor set there is no difference but for other sets (sets that are not bounded or are not closed) there is a difference.

Definition 6.30: Let E be a set of real numbers. Then E is said to have *measure zero* if for every $\varepsilon>0$ there is a finite or infinite sequence

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots$$

of open intervals covering the set E so that

$$\sum_{k=1}^{\infty}(b_k - a_k) \leq \varepsilon.$$

Note. In the definition of measure zero sets is there a change if we insist on an *infinite* sequence of intervals, disallowing finite sequences? Suppose that the sequence

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots(a_N,b_N)$$

of open intervals covers the set E so that

$$\sum_{k=1}^{N}(b_k - a_k) < \varepsilon/2.$$

Then to satisfy the definition we could add in some further intervals that do not amount in length to more than $\varepsilon/2$. For example, take

$$(a_{N+p},b_{N+p}) = (0,\varepsilon/2^{p+1})$$

for $p = 1,2,3,\ldots$. Then

$$\sum_{k=1}^{\infty}(b_k - a_k) = \sum_{k=1}^{N}(b_k - a_k) + \sum_{p=1}^{\infty}\varepsilon/2^{p+1} < \varepsilon.$$

Thus the definition would not be changed if we had required infinite coverings.

Here are some examples of sets of measure zero.

Example 6.31: Every finite set has measure zero. The empty set is easily handled. If

$$E = \{x_1, x_2, \ldots, x_N\}$$

and $\varepsilon > 0$, then the sequence of intervals

$$\left(x_i - \frac{\varepsilon}{2N}, x_i + \frac{\varepsilon}{2N}\right) \quad i = 1,2,3,\ldots,N$$

covers the set E and the sum of all the lengths is ε. ◀

Example 6.32: Every infinite, countable set has measure zero. If

$$E = \{x_1, x_2, \ldots\}$$

and $\varepsilon > 0$, then the sequence of intervals

$$\left(x_i - \frac{\varepsilon}{2^{i+1}}, x_i + \frac{\varepsilon}{2^{i+1}}\right) \quad i = 1,2,3,\ldots$$

covers the set E and, since

$$\sum_{k=1}^{\infty} 2\left(\frac{\varepsilon}{2^{k+1}}\right) = \sum_{k=1}\varepsilon 2^{-k} = \varepsilon,$$

the sum of all the lengths is ε. ◀

Example 6.33: The Cantor set has measure zero. Let $\varepsilon > 0$. Choose n so that $(2/3)^n < \varepsilon$. Then the nth stage intervals in the construction of the Cantor set give us 2^n closed intervals each of length $(1/3)^n$. This covers the Cantor set with 2^n closed intervals of total length $(2/3)^n$, which is less than ε. If the closed intervals trouble you (the definition requires open intervals), see Exercise 6.8.13 or argue as follows. Since $(2/3)^n < \varepsilon$ there is a positive number δ so that

$$(2/3)^n + \delta < \varepsilon.$$

Enlarge each of the closed intervals to form a slightly larger open interval, but change the length of each only enough so that the sum of the lengths of all the 2^n closed intervals does not increase by more than δ. The resulting collection of open intervals also covers the Cantor set, and the sum of the length of these intervals is less than ε. ◀

One of the most fundamental of the properties of sets having measure zero is how sequences of such sets combine. We recall that the union of any sequence of countable sets is also countable. We now prove that the union of any sequence of measure zero sets is also a measure zero set.

Theorem 6.34: *Let E_1, E_2, E_3, ... be a sequence of sets of measure zero. Then the set E formed by taking the union of all the sets in the sequence is also of measure zero.*

Proof. Let $\varepsilon > 0$. We shall construct a cover of E consisting of a sequence of open intervals of total length less than ε. Since E_1 has measure zero, there is a sequence of open intervals

$$(a_{11}, b_{11}), (a_{12}, b_{12}), (a_{13}, b_{13}), (a_{14}, b_{14}), \ldots$$

covering the set E_1 and so that the sum of the lengths of these intervals is smaller than $\varepsilon/2$. Since E_2 has measure zero, there is a sequence of open intervals

$$(a_{21}, b_{21}), (a_{22}, b_{22}), (a_{23}, b_{23}), (a_{24}, b_{24}), \ldots$$

covering the set E_2 and so that the sum of the lengths of these intervals is smaller than $\varepsilon/4$. In general, for each $k = 1, 2, 3, \ldots$ there is a sequence of open intervals

$$(a_{k1}, b_{k1}), (a_{k2}, b_{k2}), (a_{k3}, b_{k3}), (a_{k4}, b_{k4}), \ldots$$

covering the set E_k and so that the sum of the lengths of these intervals is smaller than $\varepsilon/2^k$. The totality of all these intervals can be arranged into a single sequence of open intervals that covers every point in the union of the sequence $\{E_k\}$. The sum of the lengths of all the intervals in the large sequence is smaller than

$$\varepsilon/2 + \varepsilon/4 + \varepsilon/8 + \cdots = \varepsilon.$$

It follows that E has measure zero. ■

Let us return to the situation for the Cantor set once again. For each $\varepsilon > 0$ we were able to choose a finite cover of open intervals with total length less than ε. This is not the case for all sets of measure zero. For example, the set of all rational numbers on the

real line is countable and hence also of measure zero. Any finite collection of intervals must fail to cover that set, in fact cannot come close to covering all rational numbers. For what sets is it possible to select finite coverings of small length? The answer is that this is possible for compact sets of measure zero.

Theorem 6.35: *Let E be a compact set of measure zero. Then for every $\varepsilon > 0$ there is a finite collection of open intervals*

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots,(a_N,b_N)$$

that covers the set E and so that

$$\sum_{k=1}^{N}(b_k-a_k)<\varepsilon.$$

Proof. Since E has measure zero, it is certainly possible to select a sequence of open intervals

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots$$

that covers the set E and so that

$$\sum_{k=1}^{\infty}(b_k-a_k)<\varepsilon.$$

But how can we reduce this collection to a finite one that also covers the set E? If you studied the Heine-Borel theorem (Theorem 4.33), then you know how.

We shall present here a proof that uses the Bolzano-Weierstrass theorem instead. We claim that we can find an integer N so that all points of E are in one of the intervals

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots,(a_N,b_N).$$

This will prove the theorem.

We prove this by contradiction. If this is not so, then for each integer $k = 1,2,3,\ldots$ we must be able to find a point $x_k \in E$ but x_k is not in any of the intervals

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots(a_k,b_k).$$

The sequence $\{x_k\}$ is bounded because E is bounded. By the Bolzano-Weierstrass theorem the sequence has a convergent subsequence $\{x_{n_j}\}$. Let z be the limit of the convergent subsequence. Since E is closed z is in E. The original sequence of intervals covers all of E and so there must be an interval (a_M,b_M) that contains z. For large values of j the points x_{n_j} also belong to (a_M,b_M). But this is impossible since x_{n_j} cannot belong to the interval (a_M,b_M) for $n_j \geq M$. Since this is a contradiction, the proof is done. ■

Exercises

6.8.1 Show that every subset of a set of measure zero also has measure zero.

6.8.2 If E has measure zero, show that the translated set

$$E+\alpha=\{x+\alpha : x\in E\}$$

also has measure zero.

6.8.3 If E has measure zero, show that the expanded set

$$cE = \{cx : x \in E\}$$

also has measure zero for any $c > 0$.

6.8.4 If E has measure zero, show that the reflected set

$$-E = \{-x : x \in E\}$$

also has measure zero.

6.8.5 Without referring to the proof of Theorem 6.34, show that the union of any two sets of measure zero also has measure zero.

6.8.6 If $E_1 \subset E_2$ and E_1 has measure zero but E_2 has not, what can you say about the set $E_2 \setminus E_1$?

6.8.7 Show that any interval (a,b) or $[a,b]$ is not of measure zero.

6.8.8 Give an example of a set that is not of measure zero and does not contain any interval $[a,b]$.

6.8.9 A careless student claims that if a set E has measure zero, then it is obviously true that the closure $\overline{E}$ must also have measure zero. Is this correct?

6.8.10 If a set E has measure zero what can you say about interior points of that set?

6.8.11 Suppose that a set E has the property that $E \cap [a,b]$ has measure zero for every compact interval $[a,b]$. Must E also have measure zero?

6.8.12 Show that the set of real numbers in the interval $[0,1]$ that do not have a 7 in their infinite decimal expansion is of measure zero.

6.8.13 In Definition 6.30 show that closed intervals may be used without changing the definition.

6.8.14 Describe completely the class of sets E with the following property: For every $\varepsilon > 0$ there is a *finite* collection of open intervals

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots,(a_N,b_N)$$

that covers the set E and so that

$$\sum_{k=1}^{N}(b_k - a_k) < \varepsilon.$$

(These sets are said to have *zero content*.)

6.8.15 Show that a set E has measure zero if and only if there is a sequence of intervals

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),(a_4,b_4),\ldots$$

so that every point in E belongs to infinitely many of the intervals and $\sum_{k=1}^{\infty}(b_k - a_k)$ converges.

6.8.16 By altering the construction of the Cantor set, construct a nowhere dense closed subset of $[0,1]$ so that the sum of the lengths of the intervals removed is not equal to 1. Will this set have measure zero?

6.9 Challenging Problems for Chapter 6

6.9.1 Show that a function is discontinuous except at the points of a first category set if and only if it is continuous at a dense set of points.

6.9.2 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Assume that for every positive number ε the sequence $\{f(n\varepsilon)\}$ converges to zero as $n \to \infty$. Prove that

$$\lim_{x\to\infty} f(x) = 0.$$

SEE NOTE 160

6.9.3 Let f_n be a sequence of continuous functions defined on an interval $[a,b]$ such that $\lim_{n\to\infty} f_n(x) = 0$ for each $x \in [a,b]$. Show that for any $\varepsilon > 0$ there is an interval $[c,d] \subset [a,b]$ and an integer N so that

$$|f_n(x)| < \varepsilon$$

for every $n \geq N$ and every $x \in [c,d]$. Show that this need not be true for $[c,d] = [a,b]$.

6.9.4 Let f_n be a sequence of continuous functions defined on an interval $[a,b]$ such that $\lim_{n\to\infty} f_n(x) = \infty$ for each $x \in [a,b]$. Show that for any $M > 0$ there is an interval $[c,d] \subset [a,b]$ and an integer N so that

$$f_n(x) > M$$

for every $n \geq N$ and every $x \in [c,d]$. Show that this need not be true for $[c,d] = [a,b]$.

Notes

[147]Exercise 6.2.9. To make this true, assume that f is onto or else show that if E is dense then $f(E)$ is dense in the set (interval) $f(\mathbb{R})$.

[148]Exercise 6.3.1. If $q_1, q_2, q_3, \ldots$ is an enumeration of the rationals, then each of the sets $\{q_i\}$, $i \in \mathbb{N}$, is nowhere dense, but

$$\bigcup_{i=1}^{n} \{q_i\} = \mathbb{Q}$$

is not nowhere dense. (Indeed it is dense.)

[149]Exercise 6.3.2. All of (a)–(e) and (h) are true. Find counterexamples for (f) and (g). The proofs that the others are true follow routinely from the definition.

[150]Exercise 6.4.1. Suppose that

$$A_n = \bigcup_{k=1}^{\infty} A_{nk}$$

with each of the sets A_{nk} nowhere dense. Then

$$\bigcup_{n=1}^{\infty}\bigcup_{k=1}^{\infty} A_{nk} = \bigcup_{n,k=1}^{\infty} A_{nk}$$

expresses that union as a first category set.

[151]Exercise 6.4.2. Let $\{B_n\}$ be a sequence of residual subsets of $\mathbb{R}$. Thus each of the sets B_n is the complement of a first category set A_n. For each n write

$$A_n = \bigcup_{k=1}^{\infty} A_{nk}$$

with each of the sets A_{nk} nowhere dense. Then

$$B_n = \mathbb{R} \setminus \bigcup_{k=1}^{\infty} A_{nk}.$$

Now use De Morgan's laws.

[152]Exercise 6.4.3. Suppose that X is residual, that is,

$$X = \mathbb{R} \setminus \bigcup_{n=1}^{\infty} Q_n$$

where each Q_n is nowhere dense. Show that for any interval $[a,b]$ there is a point in $X \cap [a,b]$ by constructing an appropriate descending sequence of closed subintervals of $[a,b]$.

[153]Exercise 6.4.4. Make sure your sets are dense but not both residual (e.g., $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$).

[154]Exercise 6.4.5. This follows, with the correct interpretation, directly from the Baire category theorem.

[155]Exercise 6.4.7. Consider the sequence

$$A_N = \{x \in [0,1] : |f_n(x)| \leq 1, \text{ all } n \geq N\}.$$

Check that

$$\bigcup_{N=1}^{\infty} A_N = [0,1].$$

[156]Exercise 6.5.7. It is clear that there must be many irrational numbers in the Cantor ternary set, since that set is uncountable and the rationals are countable. Your job is to find just one.

[157]Exercise 6.5.10. Consider $G = (0,1) \setminus C$ where C is the Cantor ternary set.

[158]Exercise 6.6.7. Often to prove a set identity such as this the best way is to start with a point x that belongs to the set on the right and then show that point must be in the set on the left. After that is successful start with a point x that belongs to the set on the left. For example, if $f(x) > \alpha$, then

$$f(x) \geq \alpha + 1/m$$

for some integer m. But

$$f_n(x) \to f(x)$$

and so there must be an integer R so that $f_n(x) > \alpha + 1/m$ for all $n \geq R$, etc.

This exercise shows how unions and intersections of sequences of open and closed sets might arise in analysis. Note that the sets

$$\{x : f_n(x) \geq \alpha + 1/m\}$$

would be closed if the functions f_n are continuous. Thus it would follow that the set

$$\{x : f(x) > \alpha\}$$

must be of type $\mathcal{F}_\sigma$. This says something interesting about a function f that is the limit of a sequence of continuous functions $\{f_n\}$.

[159]Exercise 6.7.3. You need to recall Theorem 5.60, which asserts that monotone functions have left- and right-hand limits.

[160]Exercise 6.9.2. This is from the 1964 Putnam Mathematical Competition.

Chapter 7

DIFFERENTIATION

7.1 Introduction

Calculus courses succeed in conveying an idea of what a derivative is, and the students develop many technical skills in computations of derivatives or applications of them. We shall return to the subject of derivatives but with a different objective.

Now we wish to see a little deeper and to understand the basis on which that theory develops. Much of this chapter will appear to be a review of the subject of derivatives with more attention paid to the details now and less to the applications. Some of the more advanced material will be, however, completely new.

We start at the beginning, at the rudiments of the theory of derivatives.

7.2 The Derivative

Let f be a function defined on an interval I and let x_0 and x be points of I. Consider the *difference quotient* determined by the points x_0 and x:

$$\frac{f(x)-f(x_0)}{x-x_0}, \tag{1}$$

representing the average rate of change of f on the interval with endpoints at x and x_0.

In Figure 7.1 this difference quotient represents the slope of the chord (or secant line) determined by the points $(x, f(x))$ and $(x_0, f(x_0))$. This same picture allows a physical interpretation. If $f(x)$ represents the distance a point moving on a straight line has moved from some fixed point in time x, then $f(x) - f(x_0)$ represents the (net) dis-

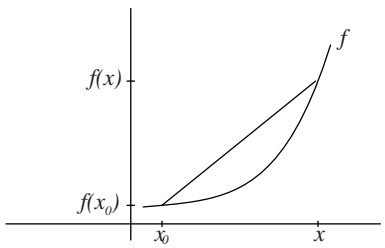


Figure 7.1. The chord determined by $(x, f(x))$ and $(x_0, f(x_0))$.

tance it has moved in the time interval $[x_0, x]$, and the difference quotient (1) represents the average velocity in that time interval.

Suppose now that we fix x_0, and allow x to approach x_0. We learn in elementary calculus that if

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists, then the limit represents the slope of the tangent line to the graph of the function f at the point $(x_0, f(x_0))$. In the setting of motion, the limit represents instantaneous velocity at time x_0.

The derivative owes its origins to these two interpretations in geometry and in the physics of motion, but now completely transcends them; the derivative finds applications in nearly every part of mathematics and the sciences.

We shall study the structure of derivatives, but with less concern for computations and applications than we would have seen in our calculus courses. Now we wish to understand the notion and see why it has the properties used in the many computations and applications of the calculus.

7.2.1 Definition of the Derivative

We begin with a familiar definition.

Definition 7.1: Let f be defined on an interval I and let $x_0 \in I$. The *derivative* of f at x_0, denoted by $f'(x_0)$, is defined as

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}, \tag{2}$$

provided either that this limit exists or is infinite. If $f'(x_0)$ is finite we say that f is *differentiable* at x_0. If f is differentiable at every point of a set $E \subset I$, we say that f is *differentiable* on E. When E is all of I, we simply say that f is a *differentiable* function.

Note. We have allowed infinite derivatives and they do play a role in many studies, but differentiable always refers to a finite derivative. Normally the phrase "a derivative exists" also means that that derivative is finite.

Example 7.2: Let $f(x) = x^2$ on $\mathbb{R}$ and let $x_0 \in \mathbb{R}$. If $x \in \mathbb{R}$, $x \neq x_0$, then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x^2 - x_0^2}{x - x_0} = \frac{(x - x_0)(x + x_0)}{(x - x_0)}.$$

Since $x \neq x_0$, the last expression equals $x + x_0$, so

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \to x_0} (x + x_0) = 2x_0,$$

establishing the formula, $f'(x_0) = 2x_0$ for the function $f(x) = x^2$. ◀

Let us take a moment to clarify the definition when the interval I contains one or both of its endpoints. Suppose $I = [a,b]$. For $x_0 = a$ (or $x_0 = b$), the limit in (2) is just a one-sided, or unilateral, limit. The function f is defined only on $[a,b]$ so we cannot consider points outside of that interval.

This brings us to another point. It can happen that a function that is *not* differentiable at a point x_0 does satisfy the requirement of (2) from one side of x_0. This means that the limit in (2) exists as $x \to x_0$ from that side. We present a formal definition.

Definition 7.3: Let f be defined on an interval I and let $x_0 \in I$. The *right-hand derivative* of f at x_0, denoted by $f'_+(x_0)$ is the limit

$$f'_+(x_0) = \lim_{x \to x_0+} \frac{f(x) - f(x_0)}{x - x_0},$$

provided that one-sided limit exists or is infinite. Similarly, the *left-hand derivative* of f at x_0, $f'_-(x_0)$, is the limit

$$f'_-(x_0) = \lim_{x \to x_0-} \frac{f(x) - f(x_0)}{x - x_0}.$$

Observe that, if x_0 is an interior point of I, then $f'(x_0)$ exists if and only if $f'_+(x_0) = f'_-(x_0)$. (See Exercise 7.2.8)

Example 7.4: Let $f(x) = |x|$ on $\mathbb{R}$. Let us consider the differentiability of f at $x_0 = 0$. The difference quotient (1) becomes

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x} = \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x < 0. \end{cases}$$

Thus

$$f'_+(0) = \lim_{x \to x_0+} \frac{|x|}{x} = 1$$

while

$$f'_-(0) = \lim_{x \to x_0-} \frac{|x|}{x} = -1.$$

The function has different right-hand and left-hand derivatives at $x_0 = 0$ so is not differentiable at $x_0 = 0$. ◀

Example 7.5: (A "trapping principle")

Let f be any function defined in a neighborhood I of zero. Suppose f satisfies the inequality $|f(x)| \leq x^2$ for all $x \in I$. Thus, the graph of f is "trapped" between the parabolas $y = x^2$ and $y = -x^2$. In particular, $f(0) = 0$. The difference quotient computed for $x_0 = 0$ becomes

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x)}{x},$$

from which we calculate

$$\left| \frac{f(x)}{x} \right| \leq \left| \frac{x^2}{x} \right| = |x|$$

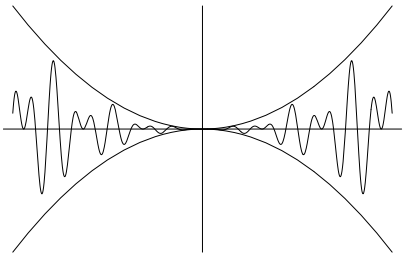


Figure 7.2. A function trapped between x^2 and $-x^2$.

so

$$\lim_{x\to 0}\left|\frac{f(x)}{x}\right| \le \lim_{x\to 0}|x| = 0.$$

Thus

$$\lim_{x\to 0}\frac{f(x)}{x} = 0.$$

As a result, $f'(0) = 0$. Figure 7.2 illustrates the principle. ◀

Higher-Order Derivatives When a function f is differentiable on I, it is possible that its derivative f' is also differentiable. When this is the case, the function $f'' = (f')'$ is called the *second derivative* of the function f. Inductively, we can define derivatives of all orders: $f^{(n+1)} = (f^{(n)})'$ (provided $f^{(n)}$ is differentiable). When n is small, it is customary to use the convenient notation f'' for $f^{(2)}$, f''' for $f^{(3)}$ etc.

Notation It is useful to have other notations for the derivative of a function f. Common notations are $\frac{df}{dx}$ and $\frac{dy}{dx}$ (when the function is expressed in the form $y = f(x)$). Another notation that is useful is Df. These alternate notations along with slight variations are useful for various calculations. You are no doubt familiar with such uses—the convenience of writing

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

when using the chain rule, or viewing D as an operator in solving linear differential equations. Notation *can* be important at times. Consider, for example, how difficult it would be to perform a simple arithmetic calculation such as the multiplication (104)(90) using Roman numerals (CIV)(XC)!

Exercises

7.2.1 You might be familiar with a slightly different formulation of the definition of derivative. If x_0 is interior to I, then for h sufficiently small, the point $x_0 + h$ is also in I. Show that expression (2) then reduces to

$$f'(x_0) = \lim_{h\to 0}\frac{f(x_0+h) - f(x_0)}{h}.$$

Repeat Examples 7.2 and 7.4 using this formulation of the derivative.

SEE NOTE 161

7.2.2 Let $c \in \mathbb{R}$. Calculate the derivatives of the functions $g(x) = c$ and $k(x) = x$ directly from the definition of derivative.

7.2.3 Check the differentiability of each of the functions below at $x_0 = 0$.

(a) $f(x) = x|x|$

(b) $f(x) = x \sin x^{-1}$ $(f(0) = 0)$

(c) $f(x) = x^2 \sin x^{-1}$ $(f(0) = 0)$

(d) $f(x) = \begin{cases} x^2, & \text{if } x \text{ rational} \\ 0, & \text{if } x \text{ irrational} \end{cases}$

7.2.4 Let $f(x) = \begin{cases} x^2, & \text{if } x \geq 0 \\ ax, & \text{if } x < 0 \end{cases}$

(a) For which values of a is f differentiable at $x = 0$?

(b) For which values of a is f continuous at $x = 0$?

(c) When f is differentiable at $x = 0$, does $f''(0)$ exist?

7.2.5 For what positive values of p is the function $f(x) = |x|^p$ differentiable at 0?

7.2.6 A function f has a *symmetric derivative* at a point if

$$f_s'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x-h)}{2h}$$

exists. Show that $f_s'(x) = f'(x)$ at any point at which the latter exists but that $f_s'(x)$ may exist even when f is not differentiable at x.

See Note 162

7.2.7 Find all points where $f(x) = \sqrt{1 - \cos x}$ is not differentiable and at those points find the one-sided derivatives.

See Note 163

7.2.8 Prove that if x_0 is an interior point of an interval I, then $f'(x_0)$ exists or is infinite if and only if $f_+'(x_0) = f_-'(x_0)$.

7.2.9 Let a function $f : \mathbb{R} \to \mathbb{R}$ be defined by setting $f(1/n) = c_n$ for $n = 1, 2, 3, \ldots$ where $\{c_n\}$ is a given sequence and elsewhere $f(x) = 0$. Find a condition on that sequence so that $f'(0)$ exists.

7.2.10 Let a function $f : \mathbb{R} \to \mathbb{R}$ be defined by setting $f(1/n^2) = c_n$ for $n = 1, 2, 3, \ldots$ where $\{c_n\}$ is a given sequence and elsewhere $f(x) = 0$. Find a condition on that sequence so that $f'(0)$ exists.

7.2.11 Give an example of a function with an infinite derivative at some point. Give an example of a function f with $f_+'(x_0) = \infty$ and $f_-'(x_0) = -\infty$ at some point x_0.

7.2.12 If $f'(x_0) > 0$ for some point x_0 in the interior of the domain of f show that there is a $\delta > 0$ so that

$$f(x) < f(x_0) < f(y)$$

whenever $x_0 - \delta < x < x_0 < y < x_0 + \delta$. Does this assert that f is increasing in the interval $(x_0 - \delta, x_0 + \delta)$?

See Note 164

7.2.13 Let f be increasing and differentiable on an interval. Does this imply that $f'(x) \geq 0$ on that interval? Does this imply that $f'(x) > 0$ on that interval?
SEE NOTE 165

7.2.14 Suppose that two functions f and g have the following properties at a point x_0: $f(x_0) = g(x_0)$ and $f(x) \leq g(x)$ for all x in an open interval containing the point x_0. If both $f'(x_0)$ and $g'(x_0)$ exist show that they must be equal. How does this compare to the trapping principle used in Example 7.5, where it seems much more is assumed about the function f.
SEE NOTE 166

7.2.15 Suppose that f is a function defined on the real line with the property that $f(x+y) = f(x)f(y)$ for all x, y. Suppose that f is differentiable at 0 and that $f'(0) = 1$. Show that f must be differentiable everywhere and that $f'(x) = f(x)$.
SEE NOTE 167

7.2.2 Differentiability and Continuity

A continuous function need not be differentiable (Example 7.4) but the converse is true. Every differentiable function is continuous.

Theorem 7.6: *Let f be defined in a neighborhood I of x_0. If f is differentiable at x_0, then f is continuous at x_0.*

Proof. It suffices to show that $\lim_{x \to x_0} (f(x) - f(x_0)) = 0$. For $x \neq x_0$,

$$f(x) - f(x_0) = \left(\frac{f(x) - f(x_0)}{x - x_0} \right) (x - x_0).$$

Now

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

and $\lim_{x \to x_0} (x - x_0) = 0$. We then obtain

$$\lim_{x \to x_0} (f(x) - f(x_0)) = (f'(x_0))(0) = 0$$

by the product rule for limits. ■

We can use this theorem in two ways. If we know that a function has a discontinuity at a point, then we know immediately that there is no derivative there. On the other hand, if we have been able to determine by some means that a function is differentiable at a point then we know automatically that the function must also be continuous at that point.

Exercises

7.2.16 Construct a function on the interval $[0, 1]$ that is continuous and is not differentiable at each point of some infinite set.
SEE NOTE 168

7.2.17 Suppose that a function has both a right-hand and a left-hand derivative at a point. What, if anything, can you conclude about the continuity of that function at that point?

7.2.18 Suppose that a function has an infinite derivative at a point. What, if anything, can you conclude about the continuity of that function at that point?
SEE NOTE 169

7.2.19 Show that if a function f has a symmetric derivative $f'_s(x_0)$ (see Exercise 7.2.6), then f must be symmetrically continuous at x_0 in the sense that $\lim_{h\to 0}[f(x_0+h)-f(x_0-h)]=0$. Must f in fact be continuous?
SEE NOTE 170

7.2.20 If $f'(x_0)=\infty$, does it follow that f must be continuous at x_0 on one side at least?

7.2.21 Find an example of an everywhere differentiable function f so that f' is not everywhere continuous.

7.2.22 Show that a function f that satisfies an inequality of the form

$$|f(x)-f(y)|\le M\sqrt{|x-y|}$$

for some constant M and all x, y must be everywhere continuous but need not be everywhere differentiable.

7.2.23 The Dirichlet function (see Section 5.2.6) is discontinuous at each rational number. By Theorem 7.6 it follows that this function has no derivative at any rational number. Does it have a derivative at any irrational number?

7.2.3 The Derivative as a Magnification

✂ *Enrichment section. May be omitted.*

We offer now one more interpretation of the derivative, this time as a magnification factor. In elementary calculus one often makes use of the geometric content of the graph of a function f. In particular, we can view the derivative in terms of slopes of tangent lines to the graph. But the graph of f is a subset of two-dimensional space, while the range of f is a subset of one-dimensional space and, as such, has some additional geometric content.

Suppose f is differentiable on an interval I, and let J be a closed sub-interval of I. The range of f on J will also be a closed interval, because f is differentiable and hence continuous on J, and continuous functions map closed intervals onto closed intervals (Exercise 5.8.2). If we compare the length $|J|$ of the interval J to the length $|f(J)|$ of the interval $f(J)$ the expression

$$\frac{|f(J)|}{|J|}$$

represents the amount that the interval J has been expanded (or contracted) under the mapping f.

For example, if $f(x)=x^2$ and $J=[2,3]$, then

$$\frac{|f(J)|}{|J|}=\frac{|[4,9]|}{|[2,3]|}=\frac{5}{1}=5.$$

Thus the interval $[2,3]$ has been expanded by f to an interval of 5 times its size. If we look only at small intervals then the derivative offers a clue to the size of the magnification factor.

If J is a sufficiently small interval having x_0 as an endpoint, then the ratio $|f(J)|/|J|$ is approximately $|f'(x_0)|$, the approximation becoming "exact in the limit." Thus $|f'(x_0)|$ can be viewed as a "magnification factor" of small intervals containing the point x_0. In our illustration with the function $f(x) = x^2$, the magnification factor at $x_0 = 2$ is $f'(2) = 4$. Small intervals about x_0 are magnified by a factor of about 4. At the other endpoint $x_0 = 3$, small intervals about x_0 are magnified by a factor of about 6.

In Exercise 7.2.26 we ask you to prove a precise statement covering the preceding discussion.

Exercises

7.2.24 What is the ratio

$$\frac{|f(J)|}{|J|}$$

for the function $f(x) = x^2$ if $J = [2, 2.001]$, $J = [2, 2.0001]$, $J = [2, 2.00001]$?

7.2.25 In this section we have interpreted $f'(x_0)$ as a magnification factor. If $f'(x_0) = 0$, does this mean that small intervals containing the point x_0 are magnified by a factor of 0 when mapped by f?

7.2.26 Let f be differentiable on an interval I and let x_0 be an interior point of I. Make precise the following statement and prove it:

$$\lim_{J \to x_0} \frac{|f(J)|}{|J|} = |f'(x_0)|.$$

7.3 Computations of Derivatives

Example 7.2 provides a calculation of the derivative of the function $f(x) = x^2$. The calculation involved direct evaluation of the limit of an appropriate difference quotient. For the function $f(x) = x^2$, this evaluation was straightforward. But limits of difference quotients can be quite complicated. You are familiar with certain rules that are useful in calculating derivatives of functions that are "built up" from functions whose derivatives are known.

In this section we review some of the calculus rules that are commonly used to compute derivatives. We need first to prove the algebraic rules: The sum rule, the product rule, and the quotient rule. Then we turn to the chain rule. Finally, we look at the power rule. Our viewpoint here is not to practice the computation of derivatives but to build up the theory of derivatives, making sure to see how it depends on work on limits that we proved earlier on.

The various rules we shall obtain in this section should be viewed as aids for computations of derivatives. An understanding of these rules is, of course, necessary for various calculations. But they in no way can substitute for an understanding of the derivative. And they might not be useful in calculating certain derivatives. (For example, derivatives of the functions of Exercise 7.2.3 cannot be calculated at $x_0 = 0$ by using these rules.)

Nonetheless, it is true that one often has a function that can be expressed in terms of several functions via the operations we considered in this section, functions whose

derivatives we know. In those cases, the techniques of this section might be useful.

7.3.1 Algebraic Rules

Functions can be combined algebraically by multiplying by constants, by addition and subtraction, by multiplication, and by division. To each of these there is a calculus rule for computing the derivative. We recall that the limit of a sum (a difference, a product, a quotient) is the sum (difference, product, quotient) of the limits. Perhaps we might have thought the same kind of rule would apply to derivatives. The derivative of the sum is indeed the sum of the derivatives, but the derivative of the product is not the product of the derivatives. Nor do quotients work in such a simple way. The reasons for the form of the various rules can be found by writing out the definition of the derivative and following through on the computations.

Theorem 7.7: *Let f and g be defined on an interval I and let $x_0 \in I$. If f and g are differentiable at x_0 then $f+g$ and fg are differentiable at x_0. If $g(x_0) \neq 0$, then f/g is differentiable at x_0. Furthermore, the following formulas are valid:*

(i) $(cf)'(x_0) = cf'(x_0)$ *for any real number c.*

(ii) $(f+g)'(x_0) = f'(x_0) + g'(x_0)$.

(iii) $(fg)'(x_0) = f(x_0)g'(x_0) + g(x_0)f'(x_0)$.

(iv) $\left(\frac{f}{g}\right)'(x_0) = \frac{g(x_0)f'(x_0) - f(x_0)g'(x_0)}{(g(x_0))^2}$ *(if $g(x_0) \neq 0$).*

Proof. Parts (i) and (ii) follow easily from the definition of the derivative and appropriate limit theorems.

To verify part (iii), let $h = fg$. Then for each $x \in I$ we have

$$h(x) - h(x_0) = f(x)[g(x) - g(x_0)] + g(x_0)[f(x) - f(x_0)]$$

so

$$\frac{h(x) - h(x_0)}{x - x_0} = f(x)\frac{g(x) - g(x_0)}{x - x_0} + g(x_0)\frac{f(x) - f(x_0)}{x - x_0}. \tag{3}$$

As $x \to x_0$, $f(x) \to f(x_0)$ since f being differentiable is also continuous. By the definition of the derivative we also know that

$$\frac{g(x) - g(x_0)}{x - x_0} \to g'(x_0)$$

and

$$\frac{f(x) - f(x_0)}{x - x_0} \to f'(x_0)$$

as $x \to x_0$. We now see from equation (3) that

$$\lim_{x \to x_0} \frac{h(x) - h(x_0)}{x - x_0} = f(x_0)g'(x_0) + g(x_0)f'(x_0),$$

verifying part (iii).

Finally, to establish part (iv) of the theorem, let $h = f/g$. Straightforward algebraic manipulations show that

$$\frac{h(x)-h(x_0)}{x-x_0} = \frac{1}{g(x)g(x_0)}\left[g(x_0)\left(\frac{f(x)-f(x_0)}{x-x_0}\right) - f(x)\left(\frac{g(x)-g(x_0)}{x-x_0}\right)\right]. \tag{4}$$

Now let $x \to x_0$. Since f and g are continuous at x_0, $f(x) \to f(x_0)$ and $g(x) \to g(x_0)$. Thus part (iv) of the theorem follows from equation (4), the definition of derivative, and basic limit theorems. ■

Example 7.8: To calculate the derivative of $h(x) = (x^3+1)^2$ we have several ways to proceed.

1. Apply the definition of derivative. You may wish to set up the difference quotient and see that a calculation of its limit is a formidable task.

2. Write $h(x) = x^6 + 2x^3 + 1$ and apply the formula $\frac{d}{dx}x^n = nx^{n-1}$ (Exercise 7.3.5) and the rule for sums. Thus we get

$$h'(x) = 6x^5 + 6x^2.$$

3. Use the product rule to obtain

$$h'(x) = (x^3+1)\frac{d}{dx}(x^3+1) + (x^3+1)\frac{d}{dx}(x^3+1).$$

Then, again, use the formula $\frac{d}{dx}x^n = nx^{n-1}$ and the rule for sums to continue:

$$h'(x) = (x^3+1)3x^2 + (x^3+1)3x^2 = 6x^5 + 6x^2.$$

◀

Exercises

7.3.1 Give the details needed in the proof of Theorem 7.7 for the sum rule for derivatives; that is, $(f+g)'(x_0) = f'(x_0) + g'(x_0)$.

7.3.2 The table shown in Figure fig–table2 gives the values of two functions f and g at certain points. Calculate $(f+g)'(1)$, $(fg)'(1)$ and $(f/g)'(1)$. What can you assert about $(f/g)'(3)$? Is there enough information to calculate $f''(3)$?

7.3.3 Obtain the rule

$$\frac{d}{dx}\frac{1}{f(x)} = -\frac{f'(x)}{f(x)^2}$$

from Theorem 7.7 and also directly from the definition of the derivative.

7.3.4 Obtain the rule for

$$\frac{d}{dx}(f(x))^2 = 2f(x)f'(x)$$

from Theorem 7.7 and also directly from the definition of the derivative.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	3	2	2
2	4	4	4	0
3	6	1	1	0
4	-1	0	1	1
5	2	5	3	3

Figure 7.3. Values of f and g at several points.

7.3.5 Obtain the formula

$$\frac{d}{dx}x^n = nx^{n-1}$$

for $n = 1, 2, 3, \ldots$ by induction.
SEE NOTE 171

7.3.6 State and prove a theorem that gives a formula for $f'(x_0)$ when

$$f = f_1 + f_2 + \cdots + f_n$$

and each of the functions $f_1, \ldots, f_n$ is differentiable at x_0.

7.3.7 State and prove a theorem that gives a formula for $f'(x_0)$ when

$$f = f_1 f_2 \ldots f_n$$

and each of the functions $f_1, \ldots, f_n$ is differentiable at x_0.

7.3.8 Show that

$$(fg)''(x_0) = f''(x_0)g(x_0) + 2f'(x_0)g'(x_0) + f(x_0)g''(x_0)$$

under appropriate hypotheses.

7.3.9 Extend Exercise 7.3.8 by obtaining a similar formula for $(fg)'''(x_0)$.

7.3.10 Obtain a formula for $(fg)^{(n)}(x_0)$ valid for $n = 1, 2, 3, \ldots$.
SEE NOTE 172

7.3.2 The Chain Rule

There is another, nonalgebraic, interpretation of Example 7.8 that you may recall from calculus courses.

Example 7.9: We can view the function $h(x) = (x^3+1)^2$ as a *composition* of the function $f(x) = x^3 + 1$ and $g(u) = u^2$. Thus

$$h(x) = g \circ f(x).$$

You are familiar with the *chain rule* that is useful in calculating derivatives of composite functions. In this case the calculation would lead to

$$\begin{aligned} h'(x) &= g'(f(x))f'(x) = g'(x^3+1)3x^2 \\ &= 2(x^3+1)3x^2 = 6x^5 + 6x^2. \end{aligned}$$

In elementary calculus you might have preferred to obtain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = 2(x^3+1)(3x^2) = 6x^5 + 6x^2$$

by making the substitution $u = x^3 + 1, y = u^2$. ◀

The chain rule is the familiar calculus formula

$$\frac{d}{dx}g(f(x)) = g'(f(x))f'(x)$$

for the differentiation of the composition of two functions $g \circ f$ under appropriate assumptions. Calculus students often memorize this in the form

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

by using the new variables $y = g(u)$ and $u = f(x)$.

Let us first try to see why the chain rule should work. Then we'll provide a precise statement and proof of the chain rule. Perhaps the easiest way to "see" the chain rule is by interpreting the derivative as a magnification factor.

Let f be defined in a neighborhood of x_0 and let g be defined in a neighborhood of $f(x_0)$. If f is differentiable at x_0, then f maps each small interval J containing x_0 onto an interval $f(J)$ containing $f(x_0)$ with $|f(J)|/|J|$ approximately $|f'(x_0)|$. If, also, g is differentiable at $f(x_0)$, then g will map a small interval $f(J)$ containing $f(x_0)$ onto an interval $g(f(J))$ with $|g(f(J))|/|f(J)|$ approximately $|g'(f(x_0))|$. Thus $h = g \circ f$ maps J onto the interval $h(J) = g(f(J))$ and

$$\frac{|h(J)|}{|J|} = \frac{|g(f(J))|}{|f(J)|}\frac{|f(J)|}{|J|}$$

and this is approximately equal to

$$|g'(f(x_0))||f'(x_0)|.$$

In short, the magnification factors $|f'(x_0)|$ and $|g'(f(x_0))|$ multiply to give the magnification factor $|h'(x_0)|$.

Example 7.10: Let us relate this discussion to our example $h(x) = (x^3 + 1)^2$. Here $f(x) = x^3 + 1$, $g(x) = x^2$. At $x_0 = 1$ we obtain $f(x_0) = 2$, $f'(x_0) = 3$, $g(f(x_0)) = 4$ and $g'(f(x_0)) = 4$. The function f maps small intervals about $x_0 = 1$ onto ones about three times as long, and in turn, the function g maps those intervals onto ones about four times as long, so the total magnification factor for the function $h = g \circ f$ is about 12 at $x_0 = 1$ (Fig. 7.4). ◀

Proof of the Chain Rule If we wished to formulate a proof of the chain rule based on the preceding discussion we could begin by writing

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = \left(\frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)}\right)\left(\frac{f(x) - f(x_0)}{x - x_0}\right) \tag{5}$$

which compares to our formula

$$\frac{|h(J)|}{|J|} = \frac{|g(f(J))|}{|f(J)|}\frac{|f(J)|}{|J|}.$$

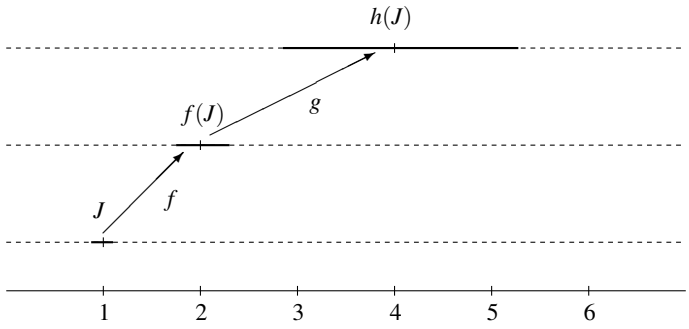


Figure 7.4. f maps J to $f(J)$ and g maps that to $h(J)$. Here $h = g \circ f$, $x_0 = 1$, and $J = [.9, 1.1]$.

If we let $x \to x_0$ in (5), we would expect to get the desired result

$$(g \circ f)'(x_0) = g'(f(x_0))f'(x_0).$$

And this argument would be valid if f were, for example, increasing. But in order for equation (5) to be valid, we must have $x \neq x_0$ and $f(x) \neq f(x_0)$. When computing the limit of a difference quotient, we can assume $x \neq x_0$, but we can't assume, without additional hypotheses, that if $x \neq x_0$ then $f(x) \neq f(x_0)$. Yet the chain rule applies nonetheless.

The proof is clearer if we separate these two cases. In the simpler case the function does not repeat the value $f(x_0)$ in some neighborhood of x_0. In the harder case the function repeats the value $f(x_0)$ in *every* neighborhood of x_0. Exercise 7.3.11 shows that in that case we must have $f'(x_0) = 0$ and so the chain rule reduces to showing that the composite function $g \circ f$ also has a zero derivative.

Theorem 7.11 (Chain Rule) *Let f be defined on a neighborhood U of x_0 and let g be defined on a neighborhood V of $f(x_0)$ for which*

$$f(x_0) \in f(U) \subset V.$$

Suppose f is differentiable at x_0 and g is differentiable at $f(x_0)$. Then the composite function $h = g \circ f$ is differentiable at x_0 and

$$h'(x_0) = (g \circ f)'(x_0) = g'(f(x_0))f'(x_0).$$

Proof. Consider any sequence of distinct points x_n different from x_0 and converging to x_0. If we can show that the sequence

$$S_n = \frac{g(f(x_n)) - g(f(x_0))}{x_n - x_0}$$

converges to $g'(f(x_0))f'(x_0)$ for every such sequence then we have obtained our required formula.

Note that if $f(x_n) \neq f(x_0)$, then we can write $y_n = f(x_n)$, $y_0 = f(x_0)$ and display S_n as

$$S_n = \left(\frac{g(y_n) - g(y_0)}{y_n - y_0}\right)\left(\frac{f(x_n) - f(x_0)}{x_n - x_0}\right). \tag{6}$$

Seen in this form it becomes obvious that

$$S_n \to g'(y_0)f'(x_0) = g'(f(x_0))f'(x_0)$$

except for the problem that we cannot (as we remarked before beginning our proof) assume that in all cases $f(x_n) \neq f(x_0)$.

Thus we consider two cases. In the first case we assume that for any sequence of distinct points x_n converging to x_0 there cannot be infinitely many terms with $f(x_n) = f(x_0)$. In that case the chain rule formula is evidently valid.

In the second case we assume that there does exist a sequence of distinct points x_n converging to x_0 with $f(x_n) = f(x_0)$ for infinitely many terms. In that case (Exercise 7.3.11) we must have $f'(x_0) = 0$ and so, to establish the chain rule, we need to prove that $h'(x_0) = 0$. But in this case for any sequence x_n converging to x_0 either $S_n = 0$ [when $f(x_n) = f(x_0)$] or else S_n can be written in the form of equation (6) [when $f(x_n) \neq f(x_0)$]. It is then clear that $S_n \to 0$ and the proof is complete. ■

Exercises

7.3.11 Show that if for each neighborhood U of x_0 there exists $x \in U$, $x \neq x_0$ for which $f(x) = f(x_0)$, then either $f'(x_0)$ does not exist or else $f'(x_0) = 0$.

SEE NOTE 173

7.3.12 Give an explicit example of functions f and g such that the "proof" of the chain rule based on equation (5) fails.

SEE NOTE 174

7.3.13 The heuristic discussion preceding Theorem 7.11 dealt with $|h'(x_0)|$, not with $h'(x_0)$. Explain how the signs of $f'(x_0)$ and $g'(f(x_0))$ affect the discussion. In particular, how can we modify the discussion to get the correct sign for $h'(x_0)$?

7.3.14 Most calculus texts use a proof of Theorem 7.11 based on the following ideas. Define a function G in the neighborhood V of $f(x_0)$ by

$$G(v) = \begin{cases} [g(v) - g(f(x_0))]/[v - f(x_0)], & \text{if } v \neq f(x_0) \\ g'(f(x_0)), & \text{if } v = f(x_0). \end{cases} \tag{7}$$

(a) Show that G is continuous at $f(x_0)$.

(b) Show that $G(v)(v - f(x_0)) = g(v) - g(f(x_0))$ for every $v \in V$, regardless of whether or not $f(x_0) = v$.

(c) Prove that $\lim_{x \to x_0} \frac{h(x) - h(x_0)}{x - x_0} = g'(f(x_0))f'(x_0)$.

7.3.15 State and prove a theorem that gives a formula for $f'(x_0)$ when

$$f = f_n \circ f_{n-1} \circ \cdots \circ f_2 \circ f_1.$$

(Be sure to state all the hypotheses that you need.)